Eastern Coalfields Limited
(A subsidiary of Coal India Limited)

Integrated Command Control Centre (ICCC) in ECL

TECHNICAL SPECIFICATION DOCUMENT

1. **Integrated Command and Control Centre (ICCC) software platform**

SN	Functional/Technical Parameters	Description
1.	General requirements	The Command & Control Application should be implemented and Compliance to the industry open standards based Commercial-of-the-shelf (COTS) products.
2.		Command & Control Application should support Role based access of application modules.
3.		System must provide a comprehensive API (Application Program Interface) or SDK (Software Development's Kit) to allow interfacing and integration with existing systems and future application and sensors which shall be deployed on the field.
4.		Command & Control Application should be able to normalize the data coming from different devices of same type and provide secure access to that data using data API(s) to application developers.
5.	Command & Control	Command & Control Application must use the latest application architecture models for better interoperability and performance.
6.		Proposed Command & Control Application shall have Web Security Audit certificate from a Cert-In authorized Security Auditor before deployment. VAPT shall be done post deployment.
7.		The Command & Control Application must be able to normalize the data from various data sources such as IOT (Internet of things), IT (Informational Technology).
8.		Command & Control Application shall be developed utilising the latest Microsoft .NET architecture /Java EE (Jakarta EE)/Node.js/ Spring Framework/ Django or any other suitable architecture and the backbone of the systems database will be a Microsoft SQL server or Postgre or any other suitable database.
9.		The proposed Command & Control Application should be deployable both on-prem and on cloud i.e., hybrid deployable.
10.		The proposed Command & Control Application must be published on at-least one of the leading global CSP marketplace i.e., AWS/Azure/GCP or any other MeitY empanelment Cloud Service Provider.
11.		The Command & Control Application must be able to integrate data from different sub-systems and provide a unified view of the sub-system data over visualization dashboards.
12.		The Command & Control Application must have the capability to perform collection of data from the devices or sensors through the APIs provided by the sub-systems.

13.		The server environment shall be capable of being proposed as a software only option that fully supports Microsoft cloud and virtualised environments to enable optional installation into a separate server location / data centre provided by others.
14.		The Command & Control Application must be compatible to work on web browsers.
15.		Must have capabilities to achieve load balancing and high availability using load balancer.
16.		ICCC shall support access to ICCC platform from multiple locations through internet/MPLS VPN/mobile data/Managed Bandwidth Services.
17.		Multiple users at Area control room as well as other important users at 13 Different areas of ECL should have the access to common ICCC platform installed at ECL HQ through internet/MPLS VPN network/Managed Bandwidth Services where they can view, monitor and generate reports with full functionalities of ICCC related to that area with user access control.
18.		Important users at ECL HQ and other remote important users should have the access to full functionalities of ICCC with user access control
19.	Integrations	The Command & Control Application shall integrate with the following Systems/ Platforms as per the requirement of ECL through SDKs and APIs:
20.		(i) Video Management System (CCTV Surveillance System),
21.		(ii) Video Analytics
22.		(iii) Boom Barriers,
23.		(iv) Access Control Systems,
24.		(v) RFID Readers,
25.		(vi) GPS Devices
26.		(vii) Vehicle Tracking System
27.		(viii) Fire Alarms
28.		(ix) Network Management System (NMS),
29.		(x) GIS platform,
30.		(xi) GPS Devices
31.		(xii) SMS Gateway, IoT devices, Weigh bridge Applications and any other similar future application.
32.	Video Management System (CCTV Surveillance System) Integration	Command & Control Application should have the capability to be integrated with Video Management System (VMS) through SDKs and APIs shared shall have the following functionalities:
33.		View live video of Camera nearest to the Alert location in Command & Control Application.
34.		User can take snapshot and allocated with alert created which can be utilized for future reference.
35.		All Cameras should be mapped on GIS platform (if integrated) with different information regarding their status, Location address.

36.		The camera visual should be directly accessed through the icon on the map.
37.		There should be a snapshot functionality, when the operator accesses a live video camera.
38.		Zoom in/out camera functionality should be provided.
39.		Ability to perform below mentioned video controls (if applicable) from Command & Control Application by Operator:
40.		Play, slide-forward, slide-backward, pause, and specify time to play recorded video.
41.		Take a video still image (snapshot) from live or recorded video.
42.		Move PTZ camera by clicking on camera controls to the chosen location.
43.		Provide screen control in video pane to control pan, tilt and zoom PTZ camera.
44.	Video Analytics Integration	The Command & Control Application should have the capability to be integrated with Video Analytics Platform through SDKs and APIs shared by their provider and shall have the following functionalities:
45.		Alerts will be generated in Command & Control Application in case of any information received from the Video Analytics platform.
46.		Geo-spatial locations of all the incidents will be plotted on GIS Map.
47.		All the information received from integrated Video Analytics Platform will be visible on Command & Control Application.
48.	Vehicle Tracking System Integration	The Command & Control Application shall have the following functionality with respect to Vehicle Tracking Integration:
49.		Track latitude and longitude or geospatial location of resources (ex. People/ vehicles.) if associated with location-based device.
50.		Enable multiple resources to be simultaneously tracked.
51.		Enable resource locations to be displayed on the map in Command & Control Application.
52.		Enable display of recent movements of tracked object on a map.
53.		Enable display of any vehicle/resource on a map with GPS coordinates.
54.		Enable search and find tracked objects by name.
55.		Operator should be able to get Vehicle's attributes by clicking on the icon of vehicle on GIS map.
56.	GIS Map Integration	Command & Control Application should be integrated with GIS map which is to be provided by the bidder at its own cost.
57.		All alerts should be mapped in GIS map
58.		Integration application should support for receiving different types of alerts.
59.		GIS map should have all basic features of creating any point of interest point or affected area.
60.		GIS map should have plotted all Cameras/Sensors deployed across the city.

61.		GIS map should enable search on Map for any alert by type or subtype.
62.		Distance measuring tool should be part of GIS panel.
63.		On receiving alerts GIS map should zoom on alert location.
64.		Clicking on any alerts in GIS map should allow to acknowledge or close the alert. Same should be possible from alert panel.
65.		Alert Panel and GIS panel should be in sync and support multi-screen.
66.		Clicking on any Alert should show nearest Cameras, from where live viewing of Camera should support.
67.		Operator should be able to take Snapshot of Live streaming and save image.
68.		Operator should have capability to Create alerts when viewing something on Camera and attaching the snapshot taken.
69.		All alerts should be mapped to various SOP, where authorities should be informed through SMS or Email.
70.		Operator should be able to draw a free hand shape on GIS map and should be able to perform following functionalities:
71.		• Alert list should be populated in a window which are generated in area within the drawn shape on GIS Map.
72.		• Operator Should be able to perform certain action on listed Alerts like close the alert, assign nearest responder and execute SOP.
73.		• Responder list should be populated in a window which are available in area within the drawn shape on GIS Map.
74.		• Operator should be able to Send the Message to selected Responders.
75.		• Operator should able to see health status of all sensors which are available in area within the drawn shape on GIS Map.
76.		Operator can assign to nearest field responder from drag and drop option from GIS Map as well as assigned from Alert panel.
77.		Platform shall have Map view widgets with support of different base maps and additional map layers imposed on the base map.
78.		Platform shall support open-source base map as well as should be able to support major map providers such as HERE, Tom-tom, Google Maps, ESRI etc.
79.		Platform shall be able to integrate and visualize map layers provided by Client for different layers such as administrative boundaries, Road Networks, Hospital/Fire Stations, Landmarks etc. These layers can be contextually enabled or disabled by the operator for situational awareness.
80.		Platform shall support the Map functionalities such as zoom-in, zoom-out, re-centre, search, filter, area selection, zoom to marker level, multi-select, selection of a specific element to see more details of the same, change of the base map, selection/deselection of layer maps for visualization.

81.		Platform shall visualize on the map different vehicles, IoT Devices & Sensors, CCTV Cameras and other IoT Elements. These layers shall show contextual information on the icons when mouse-over and have different colour of icons for easy representation of the state of the IoT elements. A summary panel shall show the legends and the corresponding counts.
82.		Platform shall have capability to show heatmap as additional map layer on top of pre-selected map layers.
83.	Alert Management	The Command & Control Application shall support the following Alert Management functionality:
84.		Operator can access complete information related with Alert.
85.		Operator should be able to be assigned nearest responder to an alert from the alert Panel itself
86.		Should support Geospatial rendering of Alert information. Platform must have the ability to show the alerts over the map view so that there is a sense of awareness amongst the platform users where the alert has occurred.
87.		Should provide facility to capture critical information such as location, name, status, time of the event and be modifiable in real time by multiple authors. Events should be captured in standard formats to facilitate event correlation and reporting.
88.		Operator should be able to print audit log of an alert from the alert panel which should contained all the activities performs against the alert.
89.		Application should provide visual indication for the severity of alert, like if severity of alert is high it should show upward-arrow, if low then downward-arrow should be visible against each alert in alert panel.
90.		Platform must have differentiated colour cues for multiple criticalities of the alert that are sourced from the sub-systems or generated in the platform.
91.		Application should provide visual indication in alert panel if alert crossed escalation time.
92.		Operators shall be able to perform the following actions:
93.		• Change the Alert state.
94.		• Edit the Alert.
95.		• Change the description
96.		• Change the priority level
97.		• Attach additional entities to the Alert.
98.		• Link related Alert.
99.		• Attach a document as a URL link to the Alert.
100.		It shall be possible to query the Alert history filtering by:
101.		• Alert type
102.		• Alert state
103.		• Location
104.		• Priority
105.		• Trigger time range

106.		• Alert owner
107.		Operator should be able to change the severity of an alert from the alert panel itself after analysing the alert.
108.		Platform must have the ability to change the severity of an alert and the ability to capture the comments when change in critically is performed by platform users.
109.		Platform users must have the capability to create manual events from the alert Management module by providing appropriate domain, alert type, description, location details and image/video for proof of alert.
110.		Operator should have clear visual indication that alert have an attachment or not.
111.		Platform users must have the ability to comment and collaborate on any event that is visible and provide more evidence, if necessary, in the form of image or video.
112.		Alert attachments should be accessible from the alert panel and user should not be required to open the complete alert to view the alert attachments.
113.		Operator should be able to short the alert based on severity, status, created time, assigned responder, alert type, alert sub type and source of alerts.
114.		Alert comments must also receive the image and video evidence submitted by the field responder, if field responder assigned for the alert.
115.		Operator should be able to clear an alert from the alert panel with mandatory remarks.
116.		Operator should be able to define up to ten escalation levels associated with the Type and Jurisdiction of an alert.
117.		Operator should be able to escalate the alert from alert panel and application should find appropriate escalation level automatically and a SMS and Email should be dispatched to qualified escalation level.
118.		Create and modify user-defined alert types. An unrestricted number of user-defined alert types shall be supported.
119.		Operator should be able to schedule the creation of alert.
120.		User should be able Set the priority level of an alert.
121.		User should be able to view the history / activity perform against the Alert.
122.		Provide the ability to group alerts by source and by type.
123.		The platform should generate Notification/ Alert messages as per the incidence / events that are received, that should be visible within the Dashboard and the Field Responder Mobile App or web portal if required.
124.		All system messages (notifications, alerts and alarms) should always be available from the Notifications View.

125.	Architecture and Data Connectors	Command & Control Application must have the ability to connect to different source systems using varied connection protocols and consume the data that can be either used for ingestion into the platform database or be directly pushed for the visualization layer to be consumed by visualization layer or be exposed as an API to be consumed by 3rd party.
126.		Command & Control Application shall be an IP enabled. All communication between the servers and other clients shall be based on standard TCP/IP protocol.
127.		Command & Control Application API/SDK should be available for third party system to use.
128.		Command & Control Application must be able to consume the data from REST API.
129.	Standard Operating Procedures (SOPs)	Ability to assign SOP to an alert type.
130.		Ability to approve or reject an SOP from getting executed for a created alert.
131.		Ability to trigger multiple tasks at the same time.
132.		Ability to trigger tasks to Field Responder mobile application through SOP.
133.		The field responder/concerned official shall enter Action Taken/ response using mobile application and report oof the same shall be available on ICCC platform.
134.	Escalation Rules	Escalation capability if the SOP is not completed within the pre-defined duration. Duration can be set by an operator and can be updated if required.
135.		Capability to create Escalation levels so that escalation hierarchy can be established when the SOPs are not executed within the pre-defined duration.
136.		Escalation rules must be region-based and can be assigned to an SOP if escalation is required.
137.		User should be able to add escalation rules up to 10 levels.
138.		Escalation rules must trigger both notification and email to the users for which the escalation level is assigned so that appropriate action can be taken by the user on the alert instance that is escalated.
139.	Visualization	Capability to create pre-defined grid-based widget system to create different visualization elements and compose it over dashboards.
140.		Capable of providing different widgets like charts, data grids, KPI, map, timeline.
141.		Able to change the themes that are allocated by the admin user for the tenant.
142.		Visualization must be intuitive that the two or more widgets should be able to communicate with each other wherein the change in one or more widgets should affect the other widgets in the dashboard.

143.		Allow widget to drilldown from one level to next level and allow other related widgets data to be automatically updated based on the selected drilldown level.
144.		Platform must be able to embed other websites/portals/standalone pages into the dashboard.
145.	Notifications	Ability to show the events from the sub-systems and be able to show in the form of notification so that the platform user can take necessary action on the alerts.
146.		Ability to show the escalation on an SOP Instance to an escalated user in the notification section.
147.		Ability to navigate from the notification section to the alert management section by perform click action on the notification.
148.		Ability to unmute or mute notification sound for the alerts.
149.		Notification section must have the ability to represent graphically from which domain/sub-system the event is received.
150.		Users must receive the notification based on the Data Access Control restriction that is set for the domain for the derived roles.
151.	Reports	Ability to schedule reports from the report admin.
152.		Ability to fetch reports based of selected parameters and can download it in Excel or PDF format.
153.		When the incident button is triggered, the footage is automatically recorded and tagged the system presents to the operator pre and post event CCTV footage for all the supported encoding and digital recording system manufacturers listed herein. This footage is then attached to the incident form and can be replayed at a touch of a button.
154.		Ability to preview the report before scheduling the report.
155.		Ability to generate the report Daily, Weekly and Monthly.
156.		Ability to pause the report schedule and resume it when required.
157.		Ability to view audit trail if the schedule got executed or failed, with appropriate transaction ids to trace in case of failures.
158.	Role based Access Control (RBAC)	In-built default Roles should be available in the Platform.
159.		Admin must have the ability to derive new roles from the default roles.
160.		Default roles must have the default features access capabilities predefined and can't be changed at any time.
161.		Ability to assign appropriate features access levels to the roles.
162.		Ability to assign appropriate domains access to the roles.
163.		Ability to assign appropriate region access to the roles.
164.		Ability to assign appropriate roles to the Users, and the users created would be able to see the data and access the features basis the role that has been assigned.
165.		Ability to grant or revoke the permission of the following functionalities to each security group role:

166.		• Create POI
167.		• Remove POI
168.		• Create Shape
169.		• Remove Shape
170.		• Cancel Alert
171.		• Print Audit Log
172.		• Escalate
173.		• Export Data Chart
174.		• Print Chart
175.		• Change Priority
176.		• Sensor location
177.		• Drag Drop Responder
178.		• Acknowledge Alert
179.		• Close Alert
180.		• View Alert
181.		• View Cancelled Alerts
182.		• Export Alerts Details
183.		• Send Email
184.		• Send SMS
185.		• View Notifications
186.		• Run Report
187.		• Save Report
188.		• Print Report
189.		• Export Report
190.	Dashboards	Command & Control Application should have KPI dashboard along with various analytics for end user.
191.		GIS Analysis: Platform should be equipped with heat map analysis along with clustering of alerts to indicate the area affected.
192.		Comparative Analysis: User should have option to compare two-time intervals along with alert type and subtype to measure the effectiveness of applying some policy.
193.		Trend Analysis: User should have visibility of all trends in system i.e., month wise, day wise, alert wise, season wise, time interval based.
194.		Dashboard shall be enabled with word cloud visual so that operator can easily identify maximum used key words.
195.		User should allow to extract event based on custom draw shape from GIS map irrespective of any boundary's limitations.
196.	Intelligent Dispatch Centre	Identify & dispatch mobile resources available nearby the Alert location.
197.	Intelligent Operator Console	Provide configurable intelligent operator console based on the jurisdiction, critical area or sensors to monitor as per situation demands for focused surveillance.
198.	Remote User Module	Supervisors remotely can access the system and monitor the alerts received, action taken status, response etc.

199.	Mass Notification System	Provide a single web-based dashboard to send notifications to target audiences using multiple communication methods including SMS, E-mail.
200.	Workflow Engine	User should be able to create the dynamic workflows with Drag and Drop ease. (as required by Customer)
201.		Workflow can be created base on action / status change of an Alert with Drag and Drop ease. (as required by Customer)
202.		User should be able to attach multiple activities with single workflow with Drag and Drop.
203.		All the activities should execute in sequential order as they are created / attached with the work flow.
204.		User should be able to attach single workflow with the Alert generated from different sub systems.
205.		User should be able to de-attach the workflow from Alert Type.
206.		User should be able to create multiple work flow without restricting a user.
207.		User Should be able to detached the Workflow from Alert Type in case it is not required.
208.	Scheduler	User Should be able to schedule an Alerts/Event for a period at specific time.
209.		User should be able to schedule to broadcast a message to individual or group users of Command & Control Application.
210.		User should able to schedule to play a file on public address system for certain period with an option to send it to individual or group of PA System (if applicable).
211.		User should able to remove any scheduled event.
212.		All scheduled event should be mapped to Calendar view in form of day, week or month.
213.	Custom Query Builder	Command & Control Application should provide user to search the system data based on the query designed by operator dynamically.
214.		User should be able to design the dynamic query using Boolean operator like AND & OR.
215.		User should be able to design dynamic query using conditional operator like IN, Greater Then, Less Then, Equal To, between and not between etc.
216.		User should be able to save these query templates for future use.
217.		Search data can be taken as Print report in form of Excel/CSV or PDF.
218.	Alert Search	Operator should be able to search alerts created based on some predefined parameters. Like From and To Date, Alert type, Alert area, alert sub type and status of alerts.
219.		Operator should be able to do analysis on searched data.
220.		Pre - defined charts should be generated automatically based on search data in 2 x 2 blocks.
221.		Operator should be able to change the type of chart in any of the panel out of four.

222.		Operator should be able to Map any chart type to any panel in 2 x 2 blocks.
223.		Operator should be able to enable or disable the level and legends of any chart panel.
224.		Operator should be able to change the X - Axis data of any chart from the search results.
225.	Correlated Alerts	Application should allow an admin operator to set the correlation parameters like correlation distance, Time etc.
226.		Application should provide correlation notification in case any newly generated alert qualifies the defined correlation criteria.
227.		Operator should be able to tag two correlated alerts manually after modifying the correlation criteria for a login session.
228.		After tagging two alerts child alert should close automatically to avoided the repetitive work for operator.
229.		Application should provide the count of alerts generated from same source in alert panel.
230.		Operator should be able to view summary of alerts generated from same source and analysed the similarity by clicking on counts of similar alerts.
231.		Operator should be able to close all similar alerts with single close action to avoid the repetitive work.
232.	Post Alert Analysis	Command & Control Application should be displaying all Alert related details by just entering the alert ID.
233.		Post Alert analysis should have complete information of Similar Alert generated within predefined time.
234.		Administrator should have the ability to change the predefined time period for displaying the similar alert.
235.		Complete history of alert action should be visible under post alert analysis.
236.		System should rate the alert execution based on certain parameter such as, closing time, priority, Action performs etc.
237.		Supervisors remotely can access the system and monitor the alerts received, action taken status, response etc.
238.		Supervisor should be able to add remarks against alerts.
239.		Supervisor should be able to give star ratings to an alert after analysing the complete details on single page.
240.	SOS Mobile App General Requirements:	The application should provide functionalities enforcing User safety.
241.		User should be able to register themselves using SOS application with minimal information like Name, Phone number, gender, age and address across ECL command area.
242.		User shall be able to create SOS alert in integrated command-and-control centre application
243.		Application should capture two pictures from front mobile camera and two from back camera automatically and attached with Created SOS alert.
244.		Number of images can be configured in application as per the requirement.

245.		Application /User shall capture 10 second video from mobile camera and attached with SOS alert created.
246.		User shall be able to attached Image, additional textual information as remarks in case SOS alert is active and not closed.
247.		User real time location shall be tracked from command-and-control center during active SOS alert.
248.		User shall view all the alerts generated by him at least as per the requirement.
249.		User shall save up to five contacts number within the application. In case SOS button pressed by user one predefined SMS should be delivered to saved Contacts.
250.		SOS Mobile App shall be the integral part of ICCC Application and shall not be a Third-party Application.
251.		User Application should be available for Both Android and IOS mobile users.
252.	Field Responder Mobile App General Requirements:	The Command & Control Application shall support mobile apps for various off-the-shelf smartphones and tablets. The mobile apps shall communicate with the Mobile Server of Command & Control Application over any WIFI or mobile network connection.
253.		Integration platform should have Work force mobile app.
254.		Mobile app should receive alerts from Command & Control Application.
255.		Mobile app should be used for GPS tracking of Resources.
256.		Mobile app should have the feature to update its status by acknowledge, enroute and arrive on the assigned alert.
257.		Mobile app user should submit the action taken report in form of text and should be able to attach audio, video files or images with the alert.
258.		Action taken report should be visible at integration platform.
259.		Mobile application should have the facility to create POIs on the map to update the new locations to keep the map updated.
260.		All the communication between the mobile apps and Command & Control Application will be on HTTP and also on HTTPS by adding TLS encryption (if required).
261.		The Application should have inbuilt capabilities to handle the Live Video Streaming request from Command & Control Application operator
262.		Mobile Responder user should be able to start live video streaming by accepting the request sent by Command & Control Application Operator.
263.		Command and Control Software operator should be able to view the live streaming of Mobile Responder Camera within Command & Control Application.
264.		Mobile Responder operator should be able to view the live streaming of Command & Control Application Console Web

		Camera (if Web Camera is installed on Command & Control Application Console) within Mobile responder Application.
265.		Mobile application should be available on latest versions of both Android & iOS platform and be upgradable or portable with latest updates in the application
266.	Chatbot functionality	Chatbot should be part of ICCC application and provide a platform to users to get the static as well as dynamic information about the ECL requirement
267.		User can register a complaint from Chatbot into ICCC application.
268.		User can view the status of registered complaint
269.		ICCC can broadcast to all chatbot users' relevant information as and when needed.
270.	Application Server Performance	Operator should be able to monitor following parameter of Command & Control Application servers on real time bases with in the Command & Control Application without using any external Tools.
271.		CPU performance counter of Application server deployed
272.		RAM Performance counter of Application server deployed
273.		/Sec disk transfer and free and used space of drive
274.		Lan Adaptor uses configured in Server
275.		System performance counter like running process , file write operations / sec, file control operations / sec etc.
276.		IIS performance counters like Request /sec, Request Wait Time ,Request Failed , Success Request etc.
277.		Above parameters can be customizes as per the requirement
278.	Database Performance	Operator should be able to monitor following parameter of Command & Control Application database server on real time bases within the Command & Control Application without using any external Tools.
279.		• Total Active requests on database.
280.		• Lock Request / sec.
281.		• Transactions / sec on database.
282.		• Database file size.
283.		• Processor Queue length.
284.		• User Connections.
285.		• Static Performance counters.
286.	Document Management	Document management should be integral part of the Command & Control Application
287.		User should be able to create dynamic file system / Folders of users' choice.
288.		User should be able to upload PDF, Word, Excel, CSV, JPEG, GIF etc. file into application to manage
289.		Uploaded files should be accessible based on user rights
290.		User should make secure the file by encrypting and should not be modify without decrypting the same.

291.		User Should be able to lock the file, so it cannot be download or access to delete from the file system
292.		Application should be able to maintain all the action perform on a file which include view, downloads, Encrypted, decrypted renaming etc.
293.		All the Logs should be accessible for further analytics.
294.		Version of uploaded files should be maintained in application itself.
295.		User should be able to rename the uploaded file.
296.		User should be able to see the dashboard and overview of complete the action performed by a user.
297.	SMS/ WhatsApp gateway	SMS/ WhatsApp gateway: The Solution should have the capability to alert the users through SMS and WhatsApp notification to the relevant stakeholders in real time.
298.		SMS/WhatsApp gateway will be provided by the successful bidder/service provider. The successful bidder/Service provider will arrange to provide the subscription/licenses (if any) for the same without any extra cost.
299.	Other Features	1. ICCC platform shall provide entire summary for various data (i.e. total coal dispatched, no. of trips, deviations, alarms etc.) for a particular point of interest (i.e check post, coal stock, weighbridge etc.) on GIS Map by clicking for a particular time period or instance.
300.		2. ICCC platform shall provide data insight from other sub systems for alarms/deviation instances generated from a particular subsystem for entire trip. Example: if deviation generated form VTS system than it shall provide entire trip detail from Check posts, weighbridge including CCTV footage/images, ANPR details at various POIs, coal loaded qty etc. for that vehicle. Same shall be done for all sub systems generated alarms.
301.		3. ICCC platform shall provide event analysis through cross data verification form all the subsystems.
302.		4. ICCC platform shall provide event analysis summary for vehicles for vehicles i.e. frequent/habitual offenders etc., transporter fleet, mine, area wise analysis for violations/deviations/alarms.
303.		5. User friendly/configurable layouts, dashboards
304.		6. ICCC platform shall be scalable for integration of various system in future (i.e. IOTs etc.)

2. Videowall along-with all hardware & software accessories

S.No.	**Minimum Technical Specifications**
Panel	
1	24X7 Operational Full HD 55" IPS Panel having HDMI and Display input ports with 3 x 3 panel configuration.

2	Brightness : 500 cd/m2
3	Contrast Ratio : 1000:1
4	Viewing Angle : 178 Deg or better
5	Response Time : 10ms or better
6	Bezel Width : 0.44 mm or better
7	Should have minimum USB ports : 1
8	Typical Power consumption: 225 watt or better
9	To be supplied with all front accessible Push/Pull wall mounting bracket/kit and accessories as per site requirement.
10	Certification: Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
Videowall processor	
11	Should support minimum 6 no. 4K input and 9 Full HD output
12	Should be preferably get connected with the workstations directly with HDMI/DP ports without any Converter
13	To be supplied with compatible controller hardware and video cables, accessories as per requirement and GUI based wall management software with KVM and Remote management feature Videowall processor Should support HDMI Audio Input and output.
14	Should be preferably get connected with the Panels directly with HDMI/DP ports without any Converter
15	Minimum Video Processing Data Rate: 4 Gbps
16	Controller should be an Enterprise Grade Rack mountable with IP-55 certified, redundant hot swappable PSU, Windows OS and RAID configured.
17	Should be preferably get connected with the workstations directly with HDMI/DP ports without any Converter

3. <u>AI based Video Analytics Application Software</u>

The Video Analytics shall offer a suite of analytics rules with Artificial Intelligence to provide automatic detection of a range of motion and non-motion behaviour of persons, objects and vehicles.
The provision of Video Analytics should be Strictly through Artificial Intelligence, Machine and Deep learning Technologies/solutions on video. 3D Calibrated Video Analytics with Old Image Comparison Technologies shall not be Accepted. There should be provision of 2 used cases (minimum) per camera.
Each of the video analytics use case shall be able to run on a unified Video Management Software Platform / Video Analytics Application Platform itself.

Following video analytics and features should be supported based on licensing:

(a) Camera Tampering Detection

i. The AI Based Camera Tampering Detection Analytics must automatically detect the tampering of camera when a fixed camera's Field of View is Tampered/Changed, covering the lens, severely defocus, paint spray etc.

ii. It must also Detect the Change or Orientation of Camera's Field of View, such that if the Camera's Orientation or Field of View is changed & no longer Observing the scene intended.

(b) Crowd detection

i. The AI Based Crowd Detection Module must detect crowd formations and triggers alarms when a specified number of people (capacity) or a specified percentage of people (occupancy) thresholds is reached in the selected area.

ii. Should have the Functionality to create multiple regions or Zones in a Field of view of Camera & should allow the user to Specify Crowd Thresholds Individually for each region. If within any region the Crowd Density Estimation exceeds the Threshold, then the system should raise an Alert.

iii. The System must have Configuration Options for User Defined Drawing/Setting of the Zone within the Field of view of the Cameras with Configurable Border Width & it's Colour.

iv. The System must have Configuration Options for Drawing Multi point Crowd Monitoring Zone where the Crowd Estimation & Counting is Desired. This Crowd Monitoring Zone Border's width & it's Depiction Colour must also Configurable.

v. The System must have Configuration Options for Inputting the User Defined Maximum Upper & Lower Limit Threshold

vi. The System must have Configuration Option for Selecting the Alarm Interval in Seconds

vii. It must be possible to configure Within a Single Filed of View/Camera, Multiple Monitoring Zones for Crowd Estimation & Counting can be Configured with each of them having different Crowd threshold Limits, The Different zone should be Distinguished with different Monitoring Zones Border Colours.

(c) Automatic Number Plate Recognition

i. Instant recognition of license plates for all passing vehicles.

ii. Accurate plate recognition even at high vehicle speeds.

iii. Handle varied lighting and weather conditions seamlessly

(d) Object counting or people/vehicle counting analytics.

i. Should Count the people in area to avoid overcrowding.

ii. It should give real time indication of number of visitors inside any store or building

iii. Shall Allow Users to Draw Lines & Count People Entering & Exiting in Real Time & should show the Total IN & Total OUT Count in the On-Screen Display

(e) Classification of objects like person, vehicle or animals

i. VA can perform object classification once the camera has been calibrated. Object classification is based on properties extracted from the object including object area and speed.

(f) Loaded/Unloaded condition of the vehicles

i. The system should be able to determine whether vehicles are loaded or unloaded. It accurately identifies the load status of the vehicle, ensuring proper monitoring and management of vehicle operations.

(g) Perimeter Violation

i. Should capture zoomed image of the object causing the Perimeter Breach and raises appropriate alarm.
ii. Should also stores the time duration between the pre and post event of the Perimeter Breach.
iii. Should have the Option Drawing Single/Multiple Lines along with Defining Logic for the Multiple Lines.

(h) Loitering Detection

i. The AI Based Loitering Detection Must automatically detect Person/Human that has moved continuously within the camera field of view in the Defined Zone for a configurable period.
ii. The Application must Support for Creation of Multiple Zones within the same Field of View of the Cameras.
iii. For Each Individual Zone, individual Configurable Parameters like Loitering Time & Name of the Zone should be Configurable.
iv. The Application must also Support for Automatic Enabling & Disabling of Video Analytics based on Time Schedule for each & every Camera.
v. The AI Based Loitering Detection Analytics Video Stream must show the Metadata like Bounding Box over the Person/Human, ID, Etc.

(i) Traffic Congestion Detection

i. Traffic congestion should be classified into 3 categories i.e. Low traffic volume, medium traffic congestion, Heavy traffic congestion
ii. Traffic congestion should calculate on the basis of number of vehicles passes through and average speed of vehicles at particular location in particular interval of time.
iii. In case, the data exceeds a particular limit an alarm in form of SMS, E-mail, pop-up will be sent to respected authority.
iv. Traffic congestion limit should be set manually by the user operating from remote location.

(j) Fire & Smoke Detection

i. The system should Classify a fire/Flame in real-time Video Stream & should give an Alarm upon Detection of Fire
ii. The System should Classify Smouldering Smoke on real-time Video Stream & should give an alarm upon Detection of Smoke.
iii. The Fire & Smoke Detection Video Analytics should be developed with Advanced Image Processing and Pattern Recognition Technology.
iv. The Fire & Smoke Detection Algorithm should be able to Dense & Sparse Fire & Smoke visible in the Camera's field of View.
v. The Fire & Smoke Detection Algorithm should be based on physical Characteristics of the Fire/Flame & Smoke Pattern detection within an Incredibly short time by analysing the Video Streams.

(k) PPE KIT detection

i. System will automatically detect and generate an alarm when a person is not wearing a safety helmet, safety harness, safety jacket, safety mask etc.

ii. It adheres to industry compliance and worker safety as per ECL requirement.

(l) Safety hazards detection

i. System shall automatically detect and generate alarms for Safety hazards: Human presence near speeding vehicles/HEMM etc., waterlogging, saggy electrical wires, any accidents etc.

ii. It adheres to industry compliance and worker safety as per ECL requirement.

(m) Insufficient illumination

i. The system should be able to monitor lighting conditions in various environments. It detects insufficient illumination by analysing real-time visual data, identifying areas with low-light conditions that could impact safety or operational efficiency.

(n) Accidents Alerts

i. Utilizes AI to analyse live video feeds from cameras to detect potential accidents based on abnormal vehicle movements, sudden stops, or collisions.

ii. Automatically generates alerts and notifications to relevant authorities or emergency services based on identified accident patterns, ensuring timely response and assistance

Features required in Analytics:

- Analytics can be applied in HD-1080p resolution, H.265, H.264 compression and 15fps.
- The VA shall support video from any camera supported by the VMS.
- Both offline and Real time analytics options available.
- Option to Email, Phone call/IVR and Push Notification Alarm of any Event to mobile phones. Push method should be without SMS
- Offline analytics possible in avi., mp4., ts, mpeg, h265, h264 and asf file formats (Any 2 formats)
- Offline analytics should run in batch mode to take every file from specified folders and run analytics one by one on each file automatically.
- Each VA alert transaction should include a minimum of two images and a 10 sec video as evidence within the VA platform. The VA system should also enable users to add remarks to specific alerts of interest. Furthermore, the system should support searching for alerts based on remarks to facilitate later review
- Minimum and maximum object size-based filtering. Filtering based on Object type.
- Define schedule of video analytics to run weekly, daily, day/night or according to user specified date and time.
- Option to run analytics in silent mode.
- View snapshots of each event and click on event to play recording of that time.
- Ability to virtually map devices to one another in the VMS. For example, Fixed cameras and PTZ cameras so that event from fixed cameras can be used to notify PTZ cameras to Zoom, Pan or tilt or go-to preset positions and take snapshots of events. PTZ Tracking should happen to track the intruding object.

- Reporting: Report of all events possible though bar and line charts.
- Ability such that alerts can be searched and categorized based on this information. i.e. Timestamp (date & time), Alert Name, Alert Type, Camera Name, Camera ID.
- Each of the video analytics use-case shall be structured as an independent module that can be deployed on any camera using a simple user interface utility, providing a complete visibility of the use cases and which cameras they are running on.
- The Video Analytics system shall be compatible with the latest technological advancements in the domain of computer vision and AI. Hence, it shall be able to quickly adapt to newer libraries and AI advancements. All the analytics and use-cases shall be based on advanced AI technology and shall not depend on traditional algorithms.
- Event Notifications: The result of each of the use case shall be in the form of events that contain the screenshot with other metadata describing the event, such as detected objects, timestamp, camera/video that generated the event and all other metadata representing the event from different use cases.
- Resource Management View: The User interface shall provide a list of all the resources available in the system such as computing servers and cameras. The status of each of the devices, whether they are online/offline shall also always be available.
- The algorithms powering the video intelligence system shall possess capability to operate parallel and distributed manner across a cluster of machines.
- The System shall be a real-time video analytics engine that utilizes advanced image processing algorithms to turn video into actionable intelligence. The AI based Video Analytics system shall consist of video- processing & analytics engine that works seamlessly both on saved videos or camera streams in real-time and provide events to the user based on the use-case basis. The system shall be compatible with all ONVIF compliant IP cameras with H.264/H.265 video decoding.
- It is possible to generate email or a text message to the designated recipients in case critical alerts are generated. The application shall escalate the alert to the designated users through email or a text message in case the alert is not acknowledged by the operator in a specified period.

VA should allow the VA applications to configure and run as per hourly/daily/weekly schedule. Users should be able to create various schedules to deploy the VA applications as required.

<u>Video Analytics Functional Requirements</u>

- Proposed intelligent AI Based Video Analytics Software (VAS) shall have the capability to provide various alarms & triggers and should notify the Authorities if any incidence/violation happens.
- The System shall be a real-time video analytics engine that utilizes advanced image processing algorithms & Artificial Intelligence to turn video into Actionable Intelligence.
- The system shall provide configurable detection zones and lines to detect events of interest, Detection zones define an area of interest and Detection lines define a Virtual Line instead of a region.

- The system shall facilitate creating multiple zones and lines in a single scene to trigger various alerts
- The system shall allow the configuration of applicable rules and manage them
- The system shall also enable editing the Zones and lines to the desired shape or size.
- The triggers generated by the applied rules shall provide visual indicators to identify the event. Such as a Green coloured Bounding Box changing the Bounding Box colour to Red on event
- The system shall enable detecting rules in the defined areas (zones/ lines)
- The system shall provide a functionality for configuring timelines for various events such as abandoned object, Stopped Vehicle Detection, Wrong Way Driving Detection etc.
- The system shall allow classification of different objects like Vehicles and people/Person.
- VAS should allow to add, edit, delete or disable and enable Policies.
- System shall have a sophisticated rule-based engine with powerful analytics capabilities that provides automatic event notification.

Note: - During the contract period of 6 years (1 yr. warranty + 5 years CAMC), if ECL desires to add more cameras towards video analytics in the future, the Bidder shall supply the additional licenses at the existing tendered rate.

All practical consideration should be taken for the effective working of system. For the above-mentioned analytics, the rules to be defined, wherever required, shall be done by ECL.

Analytics feature shall be decided by ECL as per requirement from below mentioned analytics:

a. Vehicle detection and classification.
b. Loaded/Unloaded status of vehicles.
c. Number plate detection.
d. ANPR (Automatic Number Plate Recognition)
e. Rule-based vehicle counting
f. Wrong-way driving detection.
g. Over-speeding detection.
h. Accident Detection.
i. Traffic congestion detection.
j. Detection of number of people present inside the truck during tare and gross weighment.
k. Detect and highlight incidents of vehicles bypassing weighment at weighbridge.
l. Detect and highlight incidents of vehicles bypassing Boom-barriers.
m. Crowd detection and miscellaneous event detection.
n. Unusual/ Anti-social activity detection: Detection of fire, weapons, objects-theft and incidents of vandalism.
o. Detect cases tampering of CCTVs (change in orientation, obfuscation of live view, change in focus etc.) or camera disconnection.
p. Detection of Safety hazards: Identify whether person is wearing safety kit as per prevalent safety norms. Also, other incidents pertaining to unsafe behavior (e.g. Human presence near speeding vehicles, water-logging, saggy electrical wires etc.) .
q. Weighbridge and Boom Barrier tampering.
r. Insufficient illumination Detection.

4. Servers for ICCC and Video Analytics

S.No.	Minimum Technical Specifications
1.	Processor: 2x Intel Xeon processor with minimum 48 cores (24 Core each), 2.4 Ghz or above and 11 Mb cache or AMD Equivalent & above
2.	RAM: Total minimum 256 GB Memory DDR4
3.	HDD Capacity for OS and VMS: OS & Application: 2 x 960 GB SSD in RAID1 configuration
4.	Network interface ports: Minimum 2x 1G and 2x 10G ports
5.	Redundant Power Supply: Dual, Hot-plug, Redundant Power Supply
6.	GPU Support: Should support minimum 3 GPUs per server for video analytics.
7.	RAID support: RAID 5/6
8.	Operating System: Microsoft Windows 2019 (or latest) or equivalent
9.	Form Factor: 2U Rack
10.	Server Management Software: The Systems Management software should provide Role-based security. Shall support Real-time out-of-band hardware performance monitoring & alerting, Agent-free monitoring, driver updates & configuration.
11.	Power & temperature: Real-time power meter, graphing, thresholds, alerts & capping with historical power counters. Temperature monitoring & graphing
12.	Pre-failure alert : Should provide predictive failure monitoring & proactive alerts of actual or impending component
13.	The Bidder should take utmost care to provide adequate server sizing for the VA analytics.
14.	Should be certified for relevant BIS standard of IS 13252(part1): 2010 or latest from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
15.	Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

5. Online UPS of 30 KVA with 2 hour backup at Full Load

S.No.	Minimum Technical Specifications
1	Rating in KVA : 30 KVA
2	Switching Technology : IGBT-PWM (Online UPS)
3	Input Voltage: AC three phase 300V-450V +/-10% 50Hz
4	Output Voltage: AC three Phase 230V +/-1% 50Hz
5	Overall efficiency : >/= 90%
6	Battery / Backup : With Battery
7	Type of Battery : SMF-VRLA Lead Acid Battery
8	Back up Time: 120 minutes or higher at Full Load
9	Battery Capacity (Ah) : 150 or higher

10	Protection : For under voltage at battery terminal, protection of over voltage, shot circuit, over load at UPS o/p terminal, over temperature protection
11	Digital display for metering : Input & Output voltage, Input & Output frequency etc.
12	Alarms Indication : Indicator for battery charging, over temperature, overloading & low battery voltage, etc.
13	Degree of Protection : IP20 or better
14	Accessories : Rack fitted with wheels / castos for batteries, cable and connectors as per requirement

6. 42 U Rack for Server

S.No.	Minimum Technical Specifications
1	Contemporary, sleek appearance, strong frame structure
2	Multiple vendor equipment compatibilities
3	Full line of accessories & a family of sizes and styles
4	Width: EIA Standard 19″ Rack Rails
5	External Width: 23.6″ – 600mm
6	Height: 78.74″ – 2,000mm – Rack Units: 42U
7	Depths: 39.37″ & 41.34″
8	Racks in other dimensions are available
9	Sidewalls, Split rear Doors
10	Baying Kits, Toolless Shelves, Casters, Bolt Down Kits, Cable Management
11	Power Strips – Vertical or Rackmount, Enclosure Monitoring System
12	Toolless Blanking Panels, Fans
13	For Indoor Use

7. Furniture Set

- Executive Chairs: 5 (ergonomic and adjustable, for staff at computer tables or key roles).
- Computer Tables: 5 (with cable management and storage options).
- Sofa Sets: 2 sets (3+1+1 configuration, for lounge or reception areas).
- Almirahs: 3 (lockable for documents and supplies).
- Visitor Chairs: 10 (minimal stackable design for guests or meetings).

S.No.	Specification	Details
i.	Type of Furniture	Modular desks, ergonomic chairs, storage units, meeting tables, etc.
ii.	Desk Dimensions	Customizable, usually modular sections that can be adjusted or reconfigured as needed.
iii.	Material	High-quality, durable materials such as laminated wood, metal, or composite materials.
iv.	Finish	Scratch-resistant, easy-to-clean surfaces with options for matte or gloss finishes.
v.	Ergonomics	Adjustable desks with sit/stand capabilities, ergonomic chairs with lumbar support, and adjustable armrests.

vi.	Cable Management	Integrated cable management systems with cable trays, grommets, and raceways.
vii.	Modularity	Components can be easily rearranged or reconfigured based on operational needs.
viii.	Storage Solutions	Built-in cabinets, drawers, overhead storage, and filing cabinets with lockable options.
ix.	Accessibility	Furniture designed for easy access and use by all staff, including ADA compliance where necessary.
x.	Comfort Features	Adjustable chair height, backrest tilt, and desk height adjustments for ergonomic comfort.
xi.	Durability	Heavy-duty construction capable of withstanding high use and wear.
xii.	Acoustic Treatment	Optional acoustic panels or soundproofing materials to reduce noise and enhance focus.
xiii.	Safety Features	Rounded edges, stable structures, and non-slip feet to prevent accidents.
xiv.	Assembly and Installation	Professional assembly and installation services included, with options for future reconfiguration.
xv.	Compliance	Meets relevant industry standards and regulations for furniture safety and quality.

8. Air Conditioner, 2Ton, 5 Star

S.No.	Specification	Details
1	Cooling Capacity	2 Ton
2	Energy Efficiency Rating (EER)	5-Star Rating
3	Compressor Type	Inverter Compressor
4	Power Consumption	Typically, around 1.5 - 1.9 kW
5	Voltage	230V / 50 Hz, Single Phase
6	Refrigerant Type	R32 / R410A (Eco-friendly refrigerants)
7	Cooling Technology	Advanced cooling with fast cooling, uniform air distribution, and humidity control
8	Air Filters	Anti-bacterial, dust filters, PM 2.5 filter
9	Noise Level	Indoor Unit: 35-45 dB, Outdoor Unit: 50-55 dB
10	Condenser Coil	Copper Coil
11	Airflow	4-way auto swing
12	Modes	Cool, Dry, Fan, Auto, Sleep, Turbo
13	Smart Features	Wi-Fi Enabled, Smart App Control, Voice Control (depending on model)
14	Remote Control	Yes, with backlit display

9. Unit Rate for future integration of different verticals with ICCC:
Charges for Unit Rate for future integration of different verticals with ICCC

10. Technical Manpower for 01 Year
Charges for Technical Manpower for 01 Year

11. 5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost

S.No.	Minimum Technical Specifications
1	Image Sensor: 1/2.8" or larger, Progressive scan, CMOS/CCD sensor
2	Pixel Resolution: 5 MP resolution (2592x1944) or (2560X1920)or better
3	Video Compression: H.265 or higher
4	Focal Length: 04 mm (+/- 2mm) to 10 mm (+/- 2 mm)
5	Streaming Capability: Quad streaming with any stream capable of H.265 Compression with minimum 25 FPS
6	Min. Illumination: Color Mode: 0.5 Lux
7	Min. Illumination: Black & White Mode: 0.01 Lux
8	Signal to Noise Ratio: 45 dB or better
9	IR Distance (inbuilt): 40 m
10	WDR: True WDR (120dB) or more
11	Network Connectivity: Ethernet, 10/100 Base T
12	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment. (As required for system working)
13	Trigger Event/Video Analytics: Video Motion Detection, Active Tampering Alarm, Trip Zone
14	Alarm Inputs / outputs: 1 input & 1 output
15	Privacy Masking: 4 zone
16	Input Voltage: PoE and 220V AC to DC adaptor
17	Ingress protection: IP67 or better
18	Vandal protection: IK10 or better
19	Regulatory Approvals/ Certifications: BIS certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
20	Meantime Between Failure (MTBF) rating of 1,00,000 hours. The OEM must submit an undertaking to this effect and to submit the test report from Govt. / NABL Accredited test labs / 3rd party test house of international repute such as UL, TUV etc.
21	Third party integration (API): ONVIF profile S (should be listed/display on ONVIF website)
22	This camera will be used as ANPR cameras “as well as surveillance” for access control system at check posts and shall be equipped with all necessary hardware, applicable software, licences for the functionality as mentioned in the bid documents to perform the automatic number plate recognition seamlessly. The bidder shall provide this camera with its applicable cost of ANPR licenses under his scope.
23	Onboard Storage:512 GB SDXC/SDHC/SD Memory recording or Better

12. 5 MP IP Bullet Camera (Motorized vari-focal lens)

S.No.	Minimum Technical Specifications
1	Image Sensor: 1/2.8" or larger, Progressive scan, CMOS/CCD sensor
2	Pixel Resolution: 5 MP resolution (2592x1944) or (2560X1920) or better
3	Video Compression: H.265 or higher

4	Focal Length: 04 mm (+/- 2mm) to 10 mm (+/- 2 mm)
5	Streaming Capability: Quad streaming with any stream capable of H.265 Compression with minimum 25 FPS
6	Min. Illumination: Color Mode: 0.5 Lux
7	Min. Illumination: Black & White Mode: 0.01 Lux
8	Signal to Noise Ratio: 45 dB or better
9	IR Distance (inbuilt): 40 m
10	WDR: True WDR (120dB) or more
11	Network Connectivity: Ethernet, 10/100 Base T
12	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment. (As required for system working)
13	Trigger Event/Video Analytics: Video Motion Detection, Active Tampering Alarm, Trip Zone
14	Alarm Inputs / outputs: 1 input & 1 output
15	Privacy Masking: 4 zone
16	Input Voltage: PoE and 220V AC to DC adaptor
17	Ingress protection: IP67 or better
18	Vandal protection: IK10 or better
19	Regulatory Approvals/ Certifications: BIS certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
20	Meantime Between Failure (MTBF) rating of 1,00,000 hours. The OEM must submit an undertaking to this effect and to submit the test report from Govt. / NABL Accredited test labs / 3rd party test house of international repute such as UL, TUV etc.
21	Third party integration (API): ONVIF profile S (should be listed/display on ONVIF website)
22	Onboard Storage:512 GB SDXC/SDHC/SD Memory recording or Better

13.5 MP IP Dome Camera

S.No.	Minimum Technical Specifications
1	Image Sensor: 1/2.8" or larger, Progressive scan, CMOS/CCD sensor
2	Pixel Resolution: 5 MP resolution (2592x1944) or (2560X1920 or better
3	Video Compression: H.265 or higher
4	Focal Length: any fixed focal length from 2.0 mm to 5.0 mm
5	Streaming Capability: Quad streaming with any stream capable of H.265 Compression with minimum 25 FPS
6	Min. Illumination: Colour Mode: 0.5 Lux
7	Min. Illumination: Black & White Mode: 0.01 Lux
8	Signal to Noise Ratio: 45 dB or better
9	IR Distance (inbuilt): 15 mtr
10	WDR: True WDR (120dB) or more.
11	Two way/Bi-directional Audio Input / Output required for sound detection and shall be equipped with inbuilt/external mic

12	Network Connectivity: Ethernet, 10/100 Base T
13	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment. (As required for system working)
14	Trigger Event/Video Analytics: Video Motion Detection, Active Tampering Alarm, Trip Zone
15	Alarm Inputs / outputs: 1 input & 1 output
16	Privacy Masking: 4 zone
17	Input Voltage: PoE and 220V AC to DC adaptor
18	Ingress protection: IP66 or better
19	Vandal protection: IK10 or better
20	Regulatory Approvals/ Certifications: BIS certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
21	Meantime Between Failure (MTBF) rating of 1,00,000 hours.
22	Third party integration (API): ONVIF profile S (should be listed/display on ONVIF website)
23	Onboard Storage:512 GB SDXC/SDHC/SD Memory recording or Better

14.2 MP IP, PTZ Camera

S.No.	Minimum Technical Specifications
1	Image Sensor: 1/2.8" or larger, Progressive scan, CMOS sensor
2	Pixel Resolution: 2MP (1920 x 1080) or more
3	Video Compression: H.265 or higher
4	Optical Zoom = 32 x
5	Streaming Capability: Quad streaming with any stream capable of H.265 Compression with minimum 25 FPS
6	Min. Illumination: Colour Mode 0.5 Lux
7	Min. Illumination: Black & White Mode: 0.01 Lux
8	IR Distance (inbuilt): 200 m
9	Signal to Noise Ratio: 45 dB or better
10	WDR: True WDR (120Db) or more
11	Privacy Masking: 4 zone
12	Presets: Minimum 150 presets
13	Two way/Bi-directional Audio Input / Output required for sound detection and shall be equipped with inbuilt/ external mic.
14	Auto Tracking.
15	Network Connectivity: Ethernet, 10/100 Base T
16	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment. (As required for system working)
17	Trigger Event/Video Analytics: Video Motion Detection, Active Tampering Alarm, Trip Zone
18	Alarm Inputs / outputs: 1 input & 1 output.
19	Input Voltage: PoE and 220V AC to DC adaptor

20	Ingress protection: IP66 or better
21	Vandal protection: IK10
22	Regulatory Approvals/ Certifications: BIS certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
23	Meantime Between Failure (MTBF) rating of 1,00,000 hours.
24	Third party integration (API): ONVIF profile S (should be listed/display on ONVIF website)
25	MAC address: MAC addresses should be registered in the name of the OEM.
26	Onboard Storage:512 GB SDXC/SDHC/SD Memory recording or Better

15. Enterprise CVMS (per Camera's license)

Charges for Enterprise CVMS license

16. NVR (4 channel) loaded with 1 TB Storage

S.No.	Minimum Technical Specifications
1	4 Nos Full HD better
2	Incoming Network Throughput: 40 Mbps or better
3	Outgoing Network Throughput: 40 Mbps or better
4	The NVR should be able to record all channels simultaneously at FHD resolution.
5	Camera Type: IP, 2 Mega Pixel Full HD (1920x1080) and 5.0 Megapixel (2592 x 1944) or (2560X1920) Cameras
6	Recording Resolution and Frame Rate: Support Full HD (1920x1080) @ 25 FPS and 5.0 Megapixel (2592 x 1944) or (2560X1920@25 FPS depending on Camera
7	Supported Image Resolution: Support SD resolutions, HD resolution, Full HD resolution and 5.0 Megapixel (2592 x 1944) or (2560X1920resolution depending on Camera
8	Operating System: Embedded Windows or Linux
9	Recording compression: MJPEG/ H.265 or higher
10	Video Playback: 4 Channels Simultaneously Full HD Playback in Realtime
11	Total Storage: Loaded with 1 TB or more with e-SATA Support
12	Drive Type: SATA (SAS 7200 rpm HDD or SSD)
13	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment.
14	Video Output/ Interface: HDMI/VGA
15	Network Interface: 1 Port of Ethernet 10/100/1000 Base T ports
16	Shall be equipped with USB ports
17	Video content analysis: Audio Detection, Motion Detection, Tampering
18	Audio IN and OUT
19	PTZ control
20	Recording Mode: Manual, Schedule (Continuous/Event), Event (Pre/Post), Motion detection, Alarms, Trigger Input etc.
21	Search and Export: Recording search by Camera, date and time. Export of video clips to USB Flash, Drives
22	System Log: Alarms, Events, Operator Log etc.

23	Regulatory Approvals/Certifications: BIS Certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
24	Meantime Between Failure (MTBF) rating of 1,00,000 hours.
25	MAC address: MAC addresses should be registered in the name of the OEM.

17. NVR (8 channel) loaded with 4 TB Storage

S.No.	Minimum Technical Specifications
1	8 Nos Full HD or 2 Nos 5.0 Megapixel (2592 x 1944) or (2560X1920) resolution or a mix of both
2	Incoming Network Throughput: 80 Mbps or better
3	Outgoing Network Throughput: 80 Mbps or better
4	The NVR should be able to record all channels simultaneously at FHD resolution.
5	Camera Type: IP, 2 Mega Pixel Full HD (1920x1080) and 5.0 Megapixel (2592 x 1944) or (2560X1920) Cameras
6	Recording Resolution and Frame Rate: Support Full HD (1920x1080) @ 25 FPS and 5.0 Megapixel (2592 x 1944) or (2560X1920) @25 FPS depending on Camera
7	Supported Image Resolution: Support SD resolutions, HD resolution, Full HD resolution and 5.0 Megapixel (2592 x 1944) or (2560X1920)resolution depending on Camera
8	Operating System: Embedded Windows or Linux
9	Recording compression: MJPEG / H.265 or higher
10	Video Playback: 8 Channels Simultaneously Full HD Playback in Realtime
11	Total Storage: Loaded with 4 TB or more with e-SATA Support
12	Drive Type: SATA/SAS 7200 rpm HDD or SSD
13	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment.
14	Video Output/ Interface: HDMI/VGA
15	Network Interface: 1 Ports of Ethernet 10/100/1000 Base T ports
16	Shall be equipped with USB ports
17	Video content analysis: Audio Detection, Motion Detection, Tampering
18	Audio IN and OUT
19	PTZ control
20	Recording Mode: Manual, Schedule (Continuous/Event), Event (Pre/Post), Motion detection, Alarms, Trigger Input etc.
21	Search and Export: Recording search by Camera, date and time. Export of video clips to USB Flash, Drives
22	System Log: Alarms, Events, Operator Log etc.
23	Regulatory Approvals/Certifications: BIS Certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
24	Meantime Between Failure (MTBF) rating of 1,00,000 hours. The OEM must submit an undertaking to this effect and to submit the test report from Govt. /

	NABL Accredited test labs / 3rd party test house of international repute such as UL, TUV etc.
25	MAC address: MAC addresses should be registered in the name of the OEM.

18. NVR (16 channel) loaded with 8 TB Storage

S.No.	Minimum Technical Specifications
1	16 Nos Full HD or 4 Nos 5.0 Megapixel (2592 x 1944) or (2560X1920) resolution or a mix of both
2	Incoming Network Throughput: 160 Mbps or better
3	Outgoing Network Throughput: 160 Mbps or better
4	The NVR should be able to record all channels simultaneously at FHD resolution.
5	Camera Type: IP, 2 Mega Pixel Full HD (1920x1080) and 5.0 Megapixel (2592 x 1944) or (2560X1920)Cameras
6	Recording Resolution and Frame Rate: Support Full HD (1920x1080) @ 25 FPS and 5.0 Megapixel (2592 x 1944) or (2560X1920)@25 FPS depending on Camera
7	Supported Image Resolution: Support SD resolutions, HD resolution, Full HD resolution and 5.0 Megapixel (2592 x 1944) or (2560X1920) resolution depending on Camera
8	Operating System: Embedded Windows or Linux
9	Recording compression: MJPEG/ H.265 or higher
10	Video Playback: 16 Channels Simultaneously Full HD Playback in Realtime
11	Total Storage: Loaded with 8 TB or more with e-SATA Support
12	Drive Type: SATA/SAS 7200 rpm HDD or SSD
13	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment.
14	Video Output/ Interface: HDMI/VGA
15	Network Interface: 1 Ports of Ethernet 10/100/1000 Base T ports
16	Shall be equipped with USB ports
17	Video content analysis: Audio Detection, Motion Detection, Tampering
18	Audio IN and OUT
19	PTZ control
20	Recording Mode: Manual, Schedule (Continuous/Event), Event (Pre/Post), Motion detection, Alarms, Trigger Input etc.
21	Search and Export: Recording search by Camera, date and time. Export of video clips to USB Flash, Drives
22	System Log: Alarms, Events, Operator Log etc.
23	Regulatory Approvals/ Certifications: BIS Certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
24	Meantime Between Failure (MTBF) rating of 1,00,000 hours.
25	MAC address: MAC addresses should be registered in the name of the OEM

19. NVR (32 channel) loaded with 16 TB Storage

S.No.	Minimum Technical Specifications

1	32 Nos Full HD or 8 Nos 5.0 Megapixel (2592 x 1944) or (2560X1920) resolution or a mix of both
2	Incoming Network Throughput: 224 Mbps or better
3	Outgoing Network Throughput: 224 Mbps or better
4	The NVR should be able to record all channels simultaneously at FHD resolution.
5	Camera Type: IP, 2 Mega Pixel Full HD (1920x1080) and 5.0 Megapixel (2592 x 1944) or (2560X1920) Cameras
6	Recording Resolution and Frame Rate: Support Full HD (1920x1080) @ 25 FPS and 5.0 Megapixel (2592 x 1944) or (2560X1920) @25 FPS depending on Camera
7	Supported Image Resolution: Support SD resolutions, HD resolution, Full HD resolution and 5.0 Megapixel (2592 x 1944) or (2560X1920) resolution depending on Camera
8	Operating System: Embedded Windows or Linux
9	Recording compression: MJPEG/ H.265 or higher
10	Video Playback: 16 Channels Simultaneously Full HD Playback in Realtime
11	Total Storage: Loaded with 16 TB or more with e-SATA Support
12	Drive Type: SATA/SAS 7200 rpm HDD or SSD
13	Network Protocols: Suitable and required network protocol stack to work Camera in TCP/IP based Ethernet network environment.
14	Video Output/ Interface: HDMI/VGA
15	Network Interface: 1 Ports of Ethernet 10/100/1000 Base T ports
16	Shall be equipped with USB ports
17	Video Output/ Interface: HDMI/VGA
18	Audio IN and OUT
19	PTZ control
20	Recording Mode: Manual, Schedule (Continuous/Event), Event (Pre/Post), Motion detection, Alarms, Trigger Input etc.
21	Search and Export: Recording search by Camera, date and time. Export of video clips to USB Flash, Drives
22	System Log: Alarms, Events, Operator Log etc.
23	Regulatory Approvals/Certifications: BIS Certification. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
24	Meantime Between Failure (MTBF) rating of 1,00,000 hours. The OEM must submit an undertaking to this effect and submit the test report from Govt./NABL Accredited test labs/3rd party test house of international repute such as UL, TUV etc.
25	MAC address: MAC addresses should be registered in the name of the OEM.

<u>Additional technical requirement for CCTV and NVR</u>

i. The "5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost" will be used as ANPR cameras "as well as surveillance" for access control system at check posts and shall be equipped with all necessary hardware, applicable software, licences for the functionality as mentioned in the bid documents to perform the automatic number plate recognition seamlessly

ii. The camera shall be fully supported by an open and published API (Application Programmers Interface), which shall provide necessary information for integration of functionality into third party applications.

iii. The overall CCTV system shall conform to the ONVIF (Open Network Video Interface Forum) specification which guarantees interoperability between network video products regardless of the manufacturer.

iv. The offered CCTV camera and NVR shall be of the same OEM.

v. OEM: should have ISO9001:2008 and ISO14001:2004 OR latest, to be submitted with ISO certificates.

vi. To ensure and provide self-declaration that:

a. While selecting OEMs of equipment, Hardware and Software (for this tender), the bidder has ensured that the Hardware, Software and Firmware (for this tender) including "Intellectual Property Rights" of Hardware and "Source Code" of Firmware & Software (including Cameras/NVR) selected are actual manufacturers and are not getting 3rd party manufacturing like branding and reselling in India through importing/trading and rebranding such equipment from Country that shares a Land Border with India.

However, any bidder/OEM from a country which shares a land border with India will be eligible to bid only if the bidder is registered with the competent authority.

b. While selecting OEMs of equipment, Hardware and Software (proposed to be supplied for this tender) the bidder has ensured that OEMs are not suspended by ONVIF, even though the products of the OEMs may still appear on the ONVIF website.

c. To ensure that the OEM selected should have the MAC addresses registered in the name of the OEM. The MAC address should not be in the name of any 3rd party. The MAC address should not be registered in the name of any 3rd Party/ company or any entity sharing land border with India until unless specifically allowed by the Government of India. OEM need to submit an undertaking to this effect

d. All the CCTV Cameras are of IP V 6.0 compliant.

vii. Meantime Between Failure (MTBF) rating of 1,00,000 hours. The OEM must submit an undertaking to this effect and to submit the test report from Govt. / NABL Accredited test labs / 3rd party test house of international repute such as UL, TUV etc. in respect of "5 MP BULLET IP Camera (Motorized vari-focal lens)"and "NVR (8 channel)" along with the bid

viii. Cyber-Security- The IP cameras offered must comply with essential requirement (ERs) for security prescribed by Meity as amended from time to time and should have security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, at the time of supply. The bidder is required to submit an undertaking along with the bid that the security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, shall be submitted at the time of supply.

ix. The ANPR system shall be integrated with the ICCC. The successful bidder has to arrange for the same at no extra cost to ECL.

<u>ANPR Application</u>

i. The ANPR application shall be installed at every check posts along with ANPR cameras to capture the vehicle entries and exists.

ii. The application must perform the following:

a. The software should scan for vehicle number plate and reflective acrylic label pasted on windshield displaying the vehicle number.

b. It should be able to provide separate labels for Vehicle Number Plate and Reflective acrylic label.

c. For each label, the application should return a textual description and confidence score for fine tuning purposes

d. The software application must sync with all other application modules to facilitate smooth functioning of the system.

iii. The system should be integrated for detection, analysis, storage & retrieval of the number Plate information.

iv. The system should provide alerts for hot listed vehicles when in range of detection

v. The system should be able to provide search features based on time, date, location, camera number and any digit / alphabet of the number plate.

vi. The system shall generate alarms for vehicles identified under hot listed category by user.

vii. The system shall allow user to view multiple views simultaneously for different locations.

viii. The system shall recognize and identify the various type of number plates

ix. The system shall automatically capture and recognize the number license plate of the vehicle

20. Monitor 32 inches

S.No.	Minimum Technical Specifications
1	Screen Size: 32 inches /80 cm or more and type Monitor/ TV
2	Aspect Ratio: 16:9 or more
3	Panel Type: In Plane Switching (IPS)
4	Resolution (Pixels): HD or more
5	Brightness:250 cd/sqm or more
6	Native Contrast Ratio (Minimum) 800: 1 or more
7	Viewing Angle (Horizontal: Vertical) (Degree) 170:170 or more
8	Response Time (millisecond) 10 ms or lower
9	HDMI Port/ VGA Portal
10	With accessories: All interface cables, tops, Table and Wall Mounts
11	Power Supply: AC 120-240V, 50/60Hz
12	Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

21. 55" LED Display

1. Type - LED display for 24X7 operation.
2. Screen Size: 55 Inches or bigger
3. Resolution: 4K UHD (3840 X 2160)
4. Picture processing HDR, HLG
5. 20W, 2 channel in built speaker or more
6. Inputs: Inputs: minimum HDMI X 3, USB port.

7. Power Supply: AC 120-240V ,50/60Hz
8. With accessories: All interface cables, tops, Table and Wall Mounts
9. USB media player, Auto Input search, Auto sleep, Auto wakeup, Auto source detect.
10. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

22. Server

1. No of Processors: 2 processors or more
2. 2nd or latest Generation Intel® Xeon® Scalable processors/ AMD EPYCTM, up to 16 cores per processor with Cache 24 MB or more.
3. 2.4 GHz processor or better
4. RAM: 128 GB or better
5. Total Storage capacity: 100 TB or more (including all bays). Populated with 16 TB internal storage
6. RAID (0/1/5/6) Controller.
7. USB 3.0 Ports and USB 2.0 Ports
8. Network Controller: Integrated Dual Port Gigabit Server Adapter
9. Power Supply: Redundant hot plug Power supply.
10. LED monitor 18.5'' with keyboard, mouse and all required accessories
11. Operating System: Microsoft Windows® Server latest edition
12. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

23. Video Storage

1. Populated with 120TB or more Usable Space from Day 1 and can be upgradable up to 160 TB with hot-swap storages.
2. Latest generation AMD/Intel scalable processors (1.7G, 8C, 11M Cache) or better
3. No of Processors: 2 processors or more
4. RAM: 32 GB or more
5. RAID (0/1/5/6) Controller.
6. Raid 5/6 configuration for HDDs for data redundancy from day 1.
7. It should have two or more RJ-45 ports that support 100 M, 1 G and 10 Gbps.
8. Microsoft Windows Storage Server Standard Edition/ Microsoft Windows Server Standard Edition or latest
9. USB 3.0 Ports and USB 2.0 Ports
10. Supported Protocol: IPv4, IPv6, TCP/IP, UDP, HTTP, HTTPS, SMTP, FTP, NTP, DNS/DDNS, DHCP, RTP/RTSP etc.
11. The Storage Server should have Failover feature in case of future needs
12. It should have AC100V ~ 240V Redundant Power Input
13. It should have adequate support and feature to add total cameras per site requirement
14. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

24. Workstation PC

1. Work station PC with all accessories, mouse, keyboard, antivirus, 21 inches LED display, UPS with batteries
2. Operating system: - Windows 10 or latest with recovery copy
3. Processors: - Intel core i7/i9/Xeon Gold/Silver/W series/AMD Ryzen 7/9 processor, base frequency 3.2GHz or more, 6C or more
4. RAM: DDR4 32 GB or more
5. Storage: 1 TB
6. Minimum Graphic Card 4 GB or more
7. All Hardware and Software to be IPv6 Compliant
8. Input devices: - Standard Keyboard, Optical mouse
9. Software (licensed): - MS Windows, anti-virus software licensed for complete CAMC period
10. Ports: USB 3.0 2 Nos. or more, USB 2.0 2 Nos. or more, 1 or more Standard Serial Port,1 or more Parallel port, Audio-IN/OUT, 02 Nos. or more Display ports and other communication ports as required to make system workable.
11. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

25. Printers

1. Resolution: 600 x 600 dpi AIR/2400 x 600 dpi enhanced
2. Printing speed: 10 pages per minute or more
3. Paper size: A4, B5, A5, LTR
4. Toner Saving Mode Yes
5. Power requirements 220-240V
6. Paper Input (standard): Multi-purpose tray
7. Print Speed black: up to 14 ppm or better
8. Print Technology: Laser
9. Connectivity: Hi-Speed USB 2.0 port or better
10. Memory: 2 MB or better
11. Supports manual duplex printing
12. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

26. Network Management System

1. Network Management System for monitoring and providing status report of active items installed in Network.
2. For effective operations and management of IT Operations, there is a need for an industry-standard Network Monitoring System (NMS). Given the expanse and scope of the project, NMS becomes very critical for IT Operations and SLA Measurement. Some of the critical aspects that need to be considered for operations of IT setup of are:
3. Network Fault Management
4. Network Performance Management
5. Centralized and unified Dashboard

6. There should be a tight integration between infrastructure metrics and logs to have the single consolidated console of Infrastructure & security events.
7. NMS (Network Management System) dashboard should show the network details, status, event information and a topology map.
8. NMS should be support to all network devices (i.e. IP based CCT\/ cameras, NVR, Switches, PC work station etc.) of CCTV surveillance network.
9. The solution should provide superior view of infrastructure health across system, networks, application and other IT Infrastructure components into a consolidated, central console.
10. The proposed solution should provide alert console with alert summary such as no. of correlated alert, network alert, server alert, application alert etc.
11. The system reports should be accessible via web browser and Reports can be published in PDF and csv format

27. Video Management Software and server based keylogger software centralized monitoring

1. **Description & features**
   a. It shall provide local monitoring as well as centralized monitoring of cameras connected to network.
   b. It shall be a highly scalable (to required numbers of cameras that can be added on a unit-by-unit basis) enterprise level software solution and shall offer a complete video surveillance solution.
   c. It shall operate on open architecture and shall require no proprietary IT hardware.
   d. It shall allow the controlling, recording, live monitoring, playback of archived video and data simultaneously.
   e. It shall be able to do video recording on any of the options like: inbuilt hard disks on the server, direct ~~a~~ttached storage ~~boxes~~ attached to servers video storage.
   f. It should be able to add, delete, and modify user roles & permission. The user with administrative rights shall create clients (users) and give access to the software client application based on predefined user access rights.
   g. Each camera's bit rate, frame rate and resolution shall be set independently from other cameras in the system, and altering these settings shall not affect the recording and display settings of other cameras.
   h. It shall allow for video to be streamed on workstation in Matrix or on a video wall.
   i. It shall provide drag & drop functions on the system and also for set up of connection between cameras and monitors connected to one workstation.
   j. It shall provide facility to search for recordings in the software by camera, date and time. If a date and time is specified, playback shall commence from that date and time. It shall be possible to playback more than one camera simultaneously.

   It shall be able to export sections of recordings to a separate folder, which can then be written to USB Flash Drives etc. to be played back at a location not connected to the network video management & recording network. The export process shall make available a player application, which can be provided with the exported video. Export shall be possible in mp4, avi and asf formats, windows media player or any other media player compatible format. Simultaneous export of multiple cameras shall also be possible.

   l. It shall provide facility to Export recording in client and remote PC with proper authentication.
   m. If software & server(s) operations are interrupted, like power disconnection and once

the software & server(s) are restarted, it shall automatically resume recording of any camera that was recording prior to the interruption.

n. It shall support network fault-tolerant recording such that if the network connection between a video management server and video recording server becomes unavailable, for example through cable breakage, network congestion or WLAN interruption, the system operation shall automatically ~~recover~~ resume when the connection is restored.
o. It shall support recording of H.265, H.264 & MJPEG video streams. It shall support recording of video and audio for all the channels.
p. It shall provide network time server function to ensure the synchronization of the video servers and the recordings.
q. It shall provide Video Analytics alerts like: Camera Tampering, Intrusion, Perimeter Violation, Object Left/Abandoned Baggage, Loitering, camera connectivity. (Crowd detection, Vehicle count, ANPR)
r. It shall provide E-map showing status of all cameras, devices and Alarms- Connects and Disconnect Status reports.
s. It shall provide a status of the available recording capacity.
t. It shall support at least 64 video streams concurrently on single screen. It shall support at least 4 monitors in one server/ workstation for displaying live video. It shall allow minimum 16 cameras to be replayed simultaneously.
u. It should be able to add IP camera, NVR, NAS etc. for central management.
v. It should support backup & restore feature automatically and manually.
w. Event management
    i. It should detect device status, video events, alarm input & host
    ii. It should be able to add, delete, modify, enable, disable alarm scheme
    iii. It shall receive all incoming events (motion detection and triggered digital input and relay output) in the system and take appropriate actions based on user-defined event/action relationships.
x. Storage management
    i. It should also support local storage
    ii. It should be able to take the backup from front-end device like NVR, storage etc.
y. Statistics
    i. It should be able to display detailed system information
    ii. It should be able to display running CPU status, Storage status
    iii. It should support health status of camera, NVR, storage etc.
z. It should support 3rd party camera using ONVIF
aa. It should also support mobile client. It should support 200 online users & unlimited offline users
bb. Shall provide client software for PC and Mobile phones to monitor cameras connected to same network or through internet.

2. Basic Functions: It shall perform the following applications simultaneously without interfering with any of the storage server operations

a. Live monitoring of video.
b. Live display of camera sequences.
c. Control of PTZ cameras through mouse and joystick, on screen PTZ control with Click and Zoom facility
d. Playback of archived video with play, pause, forward, and speed buttons.
e. Retrieval of archived video.
f. Instant replay of live video.
g. Mobile Surveillance & Remote Viewing

i. Ability to view live video on iOS and Android phones or devices.
ii. Ability to receive alerts/notifications on Mobile phones with and without SMS using Push technology.
iii. PTZ Control on Mobile App and Remote Locations
iv. VMS and Mobile App support for multiple sites spread across WAN to be controlled and viewed from central location.
v. Remote Administration over internet.

h. Use of graphical controls (maps) and alarm management.
i. Shall provide an authentication mechanism, which verifies the validity of the user.
j. Shall display all cameras attached to the system regardless of their physical location on the network.
k. Digital Zoom
l. Alarm Acknowledge
m. Shall provide management and control over the system using a standard PC mouse, keyboard and Digital keyboard.
n. Shall display all cameras, sequences and users in a logical tree.
o. Shall provide facility to drag and drop a camera from a tree of available cameras into any video tile for live viewing.
p. Shall provide facility to view the camera from a tree of available cameras into any video tile for live viewing.
q. Shall support graphical site representation (map) functionality, where digital maps are used to represent the physical location of cameras and other devices throughout facility.
r. Shall have the ability to contain hyperlinks to create a hierarchy of interlinked maps.
s. Shall provide facility to view the camera from a map into a video tile for live viewing in the same browser without opening a new browser.
t. Shall provide facility to click on an icon in a map to initiate PTZ camera preset, run PTZ pattern, view camera in an analog monitor or send an I/O stream.
u. Shall provide facility to decide the screen layout and also the cameras that shall be displayed on the monitors.
v. Shall be possible to switch the screen layout in response to an alarm.
w. Shall support text superimposing the title and date & time on the video.
x. It can support resolution 2560 x 1440 or better
y. The camera shall conform to ONVIF Profile S or latest. The offered product shall be available on ONVIF website and confirmation certificate to be provided on demand.

28.& 29 Application software at server & Client Application

The Central Server Application Software shall be installed at Central Server at ECL, HQ, with redundancy, in a centralized manner, for access control system as well as weighbridge automation system. It shall perform storage, analysis and interpretation of data collected from all check post / Mines entry & exit points, RFID data inclusive of weighment data generated from road weighbridges as well as data generating through Handheld RFID reader (with Sim based connectivity) for alerts, various reports with time stamping & location stamping.

Additionally, a client software for operation purpose shall be installed at entry exit points for Access control system as well as weighbridges for weighbridge automation system.

The software at central server shall be integrated to synchronize the data of Access control system as well as Weighbridge Automation System with each other and provide facility of working the client software in offline mode in case of connectivity failure.

Also, the system shall be integrated with the SAP-ERP server, Vehicle Tracking System and ICCC, and with the Truck Centring System being integrated with the Weighbridge Automation System at road weighbridges.

A. OVERALL SPECIFICATIONS

1. The bidder is required to develop and install an integrated solution for authorized entry/exit and weighment, for the Access Control System and Weighbridge Automation System, to be deployed at entry/exit checkpoints and road weighbridges, respectively. The solution must comply with the features and requirements outlined in the scope of work and technical specifications, with data transmission capability to central servers, along with integration with the SAP-ERP server as required by ECL. The entire system should function in integration, with any changes required during the contract period falling within the scope of the bidder, at no additional cost to ECL.

   The developed integrated software should be independent to hardware make & model and capable of integrating with presently deployed infrastructure at running road weighbridges of ECL such as digitizers, RFID readers, boom barriers, truck centering systems, CCTV cameras, etc. as well as must have capability to integrate with any other make & model hardware (to be deployed at any road weighbridges during contract period).

2. It shall have capability to define the route chart of any vehicle like one vehicle is allowed through only particular check post and have only permission to pass through specific road weighbridge and only allowed to exit through particular check post. And this feature should be user programmable and set according to as per requirement of ECL.
3. It shall store image of vehicle and license plate and enter the same into DBMS like MySql, PostgreSQL etc. or any other user specified database along with date timestamp, its location details.
4. The tag and vehicle related data shall be incorporated in the database upon registration of the vehicle to be engaged in coal transportation/sales/ other purpose and tag to be issued to the vehicle thereafter.
5. The RFID reader and ANPR cameras shall read the tag affixed to the wind screen and Number plate respectively, of the vehicle from the readable distance as mentioned in technical specification and allow the vehicle to enter by opening the boom barrier
6. **The software shall ensure the following:**
   a. The vehicle shall ply within the RFID tag validity period by recording its date and time at the entry and exit of the access point as well as weighbridge. If the validity expires the system will intercept the vehicle by not allowing the gate to be open besides raising the alarm, pop up exception report on the workstation/ desktop monitor installed in locally, Area control room/ Headquarter control room. Such exception will be neutralized by the nodal officer/ or any officer authorized (for the purpose) by GM of the area having validated his user ID in the system. The software shall have the facility to neutralized exception by entry of remarks by nodal officer and provision to allow and dis allow exception. Report of all such activity shall be logged properly &

chronologically in order for any future reference by any agency in non-editable format.

b. The software shall have time control facility to monitor the travel time of vehicles engaged in coal transportation, aiming to prevent undue delays or stoppages that could lead pilferage of coal. This will involve assessing the reasonable travel time between source and discharge points, as well as after passing through check posts and reaching to weighbridges, and vice versa. It shall generate alerts locally, at the area control room, and at the ECL, HQ control room, allowing for interception, in case unreasonable delays without prior permission are found

c. Facility should be made in software for competent authority to login to the system by his own user id / password & provide written instruction to allow or disallow any vehicle. Report of all such activity shall be logged properly & chronologically in order for any future reference in non-editable form to be inspected by any agency.

d. Ensure that weighment records and client software timestamps are synchronized with the central server. The time for recording weighments should be derived from the server rather than the client PC to maintain consistency and accuracy

e. Syncing and replication of central server database with client Workstation/PCs should be done on regular intervals.

7. It should have distributed architecture as Central server application, Area server application & Client application. This is being done to prevent outage due to network failure in the system at any point of time. In offline system the truck trips can work without network for a particular area for certain hours till the network is established. In case of network failure, local registration of RFID tags as well authentication should be available with suitable software in Check post work station and at weighbridge PC and same may be uploaded/updated at AHQ server/ECL HQ server subsequently after restoration of network as per ECL requirement. To have reliable system, it is advised to have distributed data load controlled centrally in offline mode. However, data security should be ensured during offline mode

8. The system shall be integrated with vehicle Tracking System (VTS), SAP-ERP application and ICCC. Bidder shall ensure all system & network security guidelines & approvals as per extant rules within the prevalent law/act is followed.

9. Each location shall have specific IP address as per applicable convention, user id, and path including the centralized server located at ECL, HQ.

10. Provision for customization of software as per company guidelines should be incorporated without additional cost to ECL during the contract period. Company shall reserve the right to alter the guidelines as per relevant rules or management instructions in future.

11. Vehicles without valid RFID Tag would not be allowed

12. It should have ability to blacklist/hotlist vehicles (Vehicle number via ANPR data/ RFID tag numbers) in order to block entry or exit. The blacklist so created shall be consolidated at the master control room and replicated across all check posts and all the road weighbridges to ensure that any blacklisted vehicle is unable to gain access through any check-post, at any road weighbridge and mine area location

13. It shall have following modules:

a. RFID Module: Registration, assignment, de-Tag, delete and blacklisting etc. of RFID Tag

b. Transporter Module:

i. For registration of transporters as per the format provided by the ECL and there shall be facility to upload necessary document for registration.

ii. Details of the transporting work order/sales delivery work order indicating work order no and date, party details, quantity of transportation/ sale, grade of coal, source and destination for lifting of coal, validity start and end date of the order, value of the order., etc.
iii. Facility for approval by administrator/ authorized officials for the registration.
iv. It shall have capabilities for integration with other online services/application of central government or state government.

c. Vehicle Registration Module:

i. For registration of all types of vehicles (Roadsale, internal coal carrying vehicle, ECL vehicles, others etc) as per the format provided by the ECL and there shall be facility to upload necessary document for registration
ii. Vehicle registration module should be integrated with transporter module.
iii. Vehicle registration module shall have facility to upload driver details including license upload, driver image upload etc.
iv. Vehicle registration module shall have capability to integrate with mparivahan or any other govt online services related to transport departments for fetching details.
v. This module shall also maintain whitelist, black listed vehicle, allowed, not allowed vehicles and have feature to move/declare vehicles in these lists (blacklisted, allowed etc.)

d. User Master Module:

i. Application shall have a user creation module, for multiple user logins with limited access, accessible to administrator.
ii. This module shall have option for role-based user creation based on various modules of the application as per ECL requirement.
iii. User creation module shall also have option to create user for various units of ECL i.e. Area/mine/siding/Check post/Weighbridge etc.
iv. Authentication: Only authenticated users should be able to login to the application. On link failure authentication should happen on client side.

e. Reports Module

i. It should have ability of generating report of all vehicles that have entered/Exited during a specified time slot ; List of all vehicles that are within premises (i.e, Entered but not exited yet) ; List of all vehicles that are within the premises for more than a user specified time limit; Incidence list of any particular vehicle showing how many times and when that vehicle has entered/exited the premises; Provision to generate an Alert if a vehicle that is marked a 'Not permitted' appears at the Entry/exit gate & at the road weighbridge; Provision of additional report which will be finalize in consultation with concerned users during implementation and during contract period
ii. It should have provision for generation of automatic E-mail alerts or SMS and Whatsapp alerts or both type alerts of all the exceptions generated, associated to Access Control system as well as Weighbridges Automation System, such as excess time in loading, loading beyond permissible limits, wrong grade of material entered etc. The charges for e-mail/ messages whatsapp shall be under the scope of the bidder and ECL will not pay any additional charges for the same.
iii. Application and reports must be accessible over web. It should have a web-based UI for MIS reporting and a web/desktop application for operators. The

application will be hosted at central server at ECL, HQ however the charges against the domain name shall be under the scope of the bidder.

iv. DO / transport order wise report of number of vehicles entered/ exited the mine.
v. Reports shall be generated Weighbridge wise, check post wise, mine wise, Area wise and for entire ECL for parameters and time period such as shift wise, day wise, weekly, monthly along with any specified time period. Response time for report generation shall be minimum (<30 second).
vi. Vehicle wise, transporter wise detailed trip report (having entire trip data in time, out time, weighment data along with location, image at source and destination etc.) including trip count etc.
vii. Source & destination wise /route wise report.
viii. Date wise report within range of dates.
ix. Exception reports vis a vis Action taken / neutralization report.
x. Detailed report and alerts for any system exceptions provided by authorised person of ECL with reasons mentioned.
xi. Exception/anomaly report for deviation from the process flow mentioned above including bypassing RFID and ANPR authentication.
xii. Any other Customizable Reports as per ECL requirement.
xiii. Equipment Breakdown Report, including Uptime.

B. <u>Additional Technical requirement at Client (Site) as well as Server site software:</u>

1. The system shall perform OCR (Optical Character Recognition) of the license plate characters (English alpha-numeric characters in standard fonts) at each Check-post/entry-exit.

2. Data Entry
   a. Automatic data fill on successful authentication.
   b. Automatic weight fills from weighbridge.
   c. Automatic signal lights as per settings.
   d. On successful save ability to generate ticket automatically.
   e. All data should save with operator name for audit purposes.
   f. Option to cancel ticket as per authorization rights.
   g. Ability to reset forms

3. Setup & Configuration
   a. Ability to add/update/delete Site details.
   b. Ability to add/update/delete Site area details.
   c. Ability to add/update/delete Item details.
   d. Ability to add/update/delete party/contractor details.
   e. Ability to add/update/delete Vehicle details.
   f. Ability to add/update/delete driver details.
   g. Any other master data should be dynamically handled through a master screen. No data should be hardcoded on the page.

4. Settings
   a. Ability to show/add weights in both KG and TON.
   b. Ability to dynamically define round off and decimal places for weight.
   c. Ability to dynamically define signal timings.

5. Users
   a. Ability to add/update/delete/search users.
   b. Ability to arrange users in groups.

c. Ability to active/inactivate users.
d. Ability to create admin users.

6. Security
a. Proper Authentication & Authorization should be throughout application.
b. Role based authorization on group/users should be implemented on each screen.
c. Password should be in encrypted in database.
d. Role based authorization on critical functionality like Cancel Tickets, Delete Tickets etc.
e. All Session should expire on logout.
f. Session should timeout after idle time of 20 mins.
g. Role based shutdown of machine on logout.
h. Code should handle and avoid any sort of SQL injection.

7. Reporting
a. Ability to generate reports based on – Date Range, Site, Item, Vehicle, Ticket, Reference number, Party/Contractor etc
b. Ability to generate report to be used for MIS input.
c. Ability to generate event tracking report for all events by operators on different sites.
d. All reports should be exported to pdf and excel.

8. Search & Analysis
a. Ability to search based on – Date Range, Site, Item, Vehicle, Ticket, Reference number, Party/Contractor etc
b. Ability to view comparison analysis based on items and date range with graphical representation.

9. Dashboard: Software should have dashboard showing – numbers of vehicles in, Number of vehicles out, Number of vehicles pending to be out, Total Net Weight per day for each site separately as well as together

10. Utilities
a. Ability to create database backup.
b. Ability to export passes data to excel.
c. Ability to import master data from excel.
d. Ability to purge outdated data

11. Signal Light Controller Integration: Ability to handle signal Lights manually

12. Database
a. Database should be in DBMS like MySql, PostgreSQL etc. or any other user specified database at server.
b. Database should be in DBMS like MySql, PostgreSQL etc. or any other user specified database at Client level to handle data operations offline.
c. Data should be pushed to server database whenever connection is available.
d. Data should always be pushed on full. No incomplete data should be pushed.
e. In case of link failure during data transferring rollback should be done.
f. Database should create backup each day.
g. Database should not use administrator account.
h. Database should follow normalization
i. In case of link failure during data transferring rollback should be done.

C. <u>ADDITIONAL SPECIFICATION FOR WEIGHBRIDGE AUTOMATION SYSTEM SOFTWARE</u>

1. Currently, weighbridges are installed with RFID reader/ writer to read RFID tags, Boom Barriers with controller, Traffic Signal Lights, Printers, Desktop PC, Managed POE switch,

Digitizer, CCTV cameras etc. The weighment software, integrated with Truck Centering System, using the existing resources and infrastructure, will be installed by the successful bidder in the weighbridges.

2. The weighment data (from the Digitizer) and vehicle details (through RFID tag reading) and its pictures through CCTV cameras, with time stamp shall be loaded automatically to the software after ensuring proper positioning of vehicle on the platform by truck centering system. The boom barrier and signal lights will be operated by the software without human intervention.
3. Weighment data shall be collected from weight indicator / digitizer and the same shall be incorporated into the data base.
4. The tag and vehicle related data shall be incorporated in the database upon registration of the vehicle to be engaged in coal transportation/sales and tag to be issued to the vehicle thereafter. Vehicles carrying explosives or any other material, where weighment is required, that vehicle is to be issued RFID tag and register with the RFID data base for information and record.
5. The RFID reader installed at the weighbridge shall read the tag affixed to the wind screen of the vehicle from the readable distance as mentioned in technical specification and allow the vehicle to enter the weighbridge by opening the boom barrier. The truck then will position itself properly on the platform using truck centering system, and issuance of weigh slips & truck tax invoice will be carried out having QR code. The weighment record on the server will be the same as stored in weighbridge computer. Entry & exit of the vehicle at the weighbridge indicating date & time shall be recorded and shown in data base regardless of whether the weighment was approved by the weighbridge operator or not due to excess load relative to the RLW or a shortfall in the balance quantity. The weighment record will also display the gross, tare, and net values of coal in kilograms, as the measurable unit for weighbridges is kilograms. The least count of existing weighbridges is 10 kg.
6. The software shall ensure the following:
    a. The vehicle plying between the two points and being weighed at two points i.e. source and destination (to be implemented at colliery sidings after installation of weighbridges there) of coal transportation route, the software shall facilitate to compare the net weight of coal between source and destination weighment point to ascertain the difference of weight. The difference of weight will be reported as exception. The Nodal officer shall act upon such exception after acknowledging the administrative action order issued by the concerned competent authority i.e. GM or his authorized representative of Area.
    b. The software shall store the last tare weight data, including the time and date, and display the most recent tare weight along with its corresponding time and date for each transaction of internal coal transportation.
    c. Software should have the facility to prevent trip diversion / multiple logging at weighbridge of loaded vehicle for any particular trip at discharge point / load point as the case may be. Interception to be generated & the vehicle will be seized for enquiry & further action.
    d. Software shall have the facility to generate an exception report for such vehicle which could not start transporting despite issuance of RFID Tags within a reasonable time to be decided by the management at the time of registration.
7. The system shall provide reports indicating the following:
    a. Vehicle wise transportation. Date in, time in, tare weight, date out, time out, gross weight, net weight & RLW
    b. DO wise report/ transport order wise report.

c. Weighbridge wise report
d. Area wise report
e. Grade wise report
f. Source & Destination wise /route wise report.
g. Date wise report within range of dates.
h. Exceptions report vis a vis Action taken / neutralization report along the time line date wise.

8. The reports can be used by the management for Transporter bill payment, ascertaining the balance quantity for DO/Transportation order & other management decisions besides access control & surveillance.

9. ECL downloads Sales Order from SAP-ERP (in excel format). The Sales Order has all the details of the coal to be shipped like party name, Shipping Address, Rates, Various Taxes, Grade of coal, Validity Date, Quantity etc. This SO information should be directly imported (from the excel file) into weighment application software or from an API link of SAP-ERP system (or in some other manner, as desired by ECL). Weighment is to be done by the weighbridge software which is in turn also integrated with CCTV system. Against each weighment an API call is to be made to the SAP-ERP system and on receiving success message from SAP-ERP weighment is to be saved.

Note: - The format of SO & all the necessary details shall be shared with the bidder during execution of the project (if required). The successful bidder shall willingly comply and provide a weighment software as per the prevailing requirements of ECL.

10. The SOs will be assigned to each weigh-bridge. Coal should be allowed to be lifted only from authorized weighbridges for a particular SO.

11. During 1st weighment (i.e., tare weight), driver comes down from the vehicle and operator locks the tare weight. CCTV will capture the images of trucks from front of the vehicle, rear of the vehicle and top of the vehicle

12. Against each 2nd weighment an API call is to be made to the SAP-ERP system and only on receiving success message, weighment information along with CCTV images will be printed & generated on the acknowledgment slip.

13. The application must have the following features:

a. Ability to restrict manual entry of vehicle nos. Only RFID Tags read nos. is to be auto-populated on the screen.
b. Two consecutive vehicles should not be allowed to be weighed till the digitizer goes to zero.
c. Flexibility setup time difference between weighment of same vehicles after a certain time-gap only. The average time taken by the vehicle to reach and come back from a particular destination along a particular route should be considered in calculating the time-difference.
d. Disallow weighment of vehicles if – vehicle is blacklisted, unauthorized entry in mines, vehicle appears for weighment before scheduled time difference.

14. Vehicles with expired SO validity status should not be allowed to enter or get weighed.

15. Vehicles with lapsed SO (i.e. where the allotted quantity has been lifted) should not be allowed to get weighed.

16. Auto-fill entries against a particular SO (like party, grade, transporter, party) upon reading a vehicle RFID Tag at weighbridge.

17. Average of tare weight – The system should keep record of last 5 tare weights recorded of a vehicle (if required by ECL). If the latest tare weight is beyond the threshold its weighment must not be allowed. Proper error message is to be displayed. Weight exceeding RLW weight should not be allowed for weighment.

18. RFID based weighment application software should not allow a greater number of vehicles against a Sales Order than those allotted for the day. Weekly allotment details should be fetched directly from the SAP-ERP system and no manual intervention in allotment is to be allowed.

19. Posting of data from sites to central database should occur on real time.

20. Capable to generate any customized report as per the requirement of ECL.

21. Successful bidder is required to create a web based online portal which will show the consolidated weighment information of ECL like daily road dispatch area wise, weighbridge wise, consumer wise, coal type wise, error reports etc, internal coal transportation reports along with dashboard. The dashboard should be interactive. This portal should also be integrated with ICCC of ECL.

22. In order to streamline the movement of only authorised vehicles in mines area, Successful bidder is required to develop a web based online portal for vehicle registration by the concerned SO holder & transporter.

The process flow is detailed below:

a. At Present, Sales Orders are generated through SAP-ERP in ECL which have to be downloaded through importing or through API link into RFID based weighment application software.

b. Plant wise weekly allotment schedule is being prepared by concerned Area Sales Managers in SAP-ERP which is also to be imported into RFID based weighment application software or to be fetched from SAP-ERP through API link.

c. Successful bidder is required to design web based online portal in such a way that it should have the provision to fetch the master data of consumer wise Sales Orders and their allotment schedule either through data already downloaded into RFID based weighment application software or through API link directly from SAP-ERP or through master data uploading by concerned sales officials.

d. Coal Consumers should have the provision to register through OTP based authentication in the proposed web based online portal. The Proposed Portal should have provision to fetch master list of consumers (along with their details such as registered Email Id, Mobile number etc.) through API from SAP-ERP or by any other means and should also have a screen to enter consumer master details by the concerned officials. The cost associated with generating OTPs via SMS and WhatsApp, as required by ECL, will be the responsibility of the bidder. ECL will not incur any additional charges for this service.

e. Only authenticated Consumers should be allowed to login at web based online portal and the portal should display the Sales Orders along with their weekly schedule against the logged-in consumer. After that, Consumer will fill vehicle details like vehicle registration number, driver details etc., going to deployed by them against the Sales Orders and their respective allotment schedule.

f. Successful bidder is required to create separate login for area officials who will upload the details of vehicles engaged in internal coal transportation in excel format or any other format as required. Uploaded details of vehicles engaged in internal coal transportation should also be integrated with Proposed Access control system.

g. Successful bidder is required to create separate login for uploading the details of vehicles engaged in miscellaneous activities in excel format or any other format as required. Uploaded details of vehicles engaged in miscellaneous activities should also be integrated with Proposed Access control system.

h. All the hardware & software including redundancy, portal security, data integrity, network security, internet charges, SMS gateway charges etc. required for establishment of proposed web based online portal would be under scope of bidder.
i. Any web application developed by the bidder for this project should be secured using:
    i. Input validation to ensure that all user input is properly validated to prevent malicious data from being entered into the system.
    ii. Implementation of secure authentication and authorization system to ensure that only authorized users can access the application.
    iii. Use encryption to protect sensitive data, both in transit and at rest.
    iv. Implementation of secure session management to prevent session hacking and protect against other session-related attacks.
    v. Regularly patch and update to keep the web application and all underlying software and libraries up to date with the latest security patches to protect against known vulnerabilities
    vi. The performance of developed web portal should be such that when user moves from one page to other page, it shall be instantaneous changeover. (i.e. this aspect of performance would be equivalent or comparable with any good Indian banking website.) and bidder should ensure round the clock availability of portal

30.24 Port Layer 2, Managed switch

S.No.	Minimum Technical Specifications
1	Managed Switch with 24 x10/100/1000 Base-T PoE ports and additional 4 x 10G SFP+ populated ports. Minimum PoE budget should be 370Watt.
2	128 Gbps or more switching capacity
3	95 Mpps or more packet forwarding rate
4	Power input range: 100 V to 240 V AC 50/60 Hz
5	Layer 2 Features: MAC address Table: 16k or more, should support STP, RSTP, MSTP, IGMP v1/v2/v3 snooping, Support IEEE 802.1Q
6	SNMP V1, V2, V3, RMON 1,2,3,9, LLDP-MED
7	Priority queuing: Strict priority/ Weighted Round Robin (WRR)
8	VLAN, Jumbo frame 9K
9	Security: Should support ACL, MAC and IP based ACLs
10	Management: Configurable Management VLAN, Admin access control via Radius and TACACS+, Web-based graphical user interface (GUI), Switch Management: HTTP/HTTPs /SSH / DHCP Client/DHCPv6 Client/ Telnet Client/ IPv6 management
11	The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.

31.8 port unmanaged gigabyte switch

8 port copper ports unmanaged switch with standard specifications. The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.

32. Industrial 8 Port POE Switch

1. Eight 10/100/1000 RJ45 Copper POE/POE+ ports with additional 2 SFP ports with at-least 240W POE/POE+ base power budget
2. 8 priority queues, MAC address table: 8K, VLAN, support spanning tree protocol (STP 802.1d, RSTP802.1w, MSTP 802.1s), Jumbo frame support
3. Layer 2 Features: IEEE 802.1p,q,x, 802.3af,at,x,u,ab standard, LLDP, LACP
4. IGMPv1, v2, v3 snooping
5. Management: WebUI, DHCP server
6. Security: Port security, dynamic ARP inspection, IP source guard, guest VLAN, storm control - unicast, multicast, broadcast, SSH, SNMPv3, TACACS+/ RADIUS server/client, BPDU guard, Access Lists
7. Quality of Service (QoS) Ingress/ Egress rate limit
8. SNMP over IPv6, HTTP/HTTP(s), Syslog, DHCPv6, SCP/SSH v6,
9. Radius/TACACS+, NTP over IPv6, DHCPv6
10. Should have PTP / NTP protocol for time synchronization
11. Operating Temperature: 0° to 60°C
12. Relative humidity of up to 95% non-condensing
13. The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.

33. 08 port managed Switch

1. Managed Eight 10/100/1000 Mbps Base-T Port with additional 1G SFP populated ports
2. 20 Gbps or more switching capacity
3. 14 Mpps or more packet forwarding rate
4. Power input range: 100 V to 240 V AC 50/60 Hz
5. Layer 2 Features: should support STP, VLAN, Support IEEE 802.1Q, SNMP, Strict priority/ Weighted Round Robin (WRR), Jumbo frame 9K
6. Security: Should support ACL, MAC and IP based ACLs
7. Management: Configurable Management VLAN, Admin access control via Radius/TACACS+, Web-based graphical user interface (GUI)
8. IPv6 complaint
9. The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.

34. 16 port POE managed Switch

1. Managed Switch with 16 x10/100/1000 Base-T PoE ports and additional 2 x 1G SFP populated ports. Minimum PoE budget should be 100 Watt.
2. 32 Gbps or more switching capacity
3. 20 Mpps or more packet forwarding rate
4. Power input range: 100 V to 240 V AC 50/60 Hz
5. Layer 2 Features: should support STP, VLAN, Support IEEE 802.1Q, SNMP, Strict priority/ Weighted Round Robin (WRR), Jumbo frame 9K
6. Security: Should support ACL, MAC and IP based ACLs

7. Management: Configurable Management VLAN, Admin access control via Radius/TACACS+, Web-based graphical user interface (GUI)
8. IPv6 complaint
9. The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.

Additional requirement for all switches:

1. Product should be IPv6 Ready from day 1.
2. OEM Must have ISO 9001, ISO 14001/18001 Certification.
3. The quoted product should be MTCTE/ TEC ER certified. The bidder should submit MTCTE/TEC ER certificate for the offered product or any product from the offered product family/ product series against requirement of switches along with the bid.
4. Switch Should support CE/BIS, EMC: TEC/SD/DD/EMC-221/05/OCT-2016, EN / IEC 62368-1:2018 or latest, EN61000-4-2, EN61000-4-3, EN61000-4-4, EN61000-4-5, EN61000-4-6, EN61000-4-8, RoHS Directive along with documentary evidence.
5. ECL reserves the right to get the switches tested and certified for VAPT from any 3rd party authorized lab approved from Govt. of India. It will be responsibility of bidder to address & close all the vulnerabilities in defined timelines without any additional cost to ECL

35. Media Converter

To be supplied in pairs with Standard Specification with 10/100/1000 Base Ethernet port

36. CAT 6 Armoured STP cable

Armoured outdoor CAT 6 STP cable, 8 count, 0.51 mm or more diameter copper conductor with standard specifications

37. CAT 6 UTP cable

UTP outdoor CAT 6 cable, 8 count, 0.51 mm or more diameter copper conductor with standard specifications

38. Armoured Optical Fibre cable

Armoured Optical Fibre cable 6 core or more with standard specifications. The offered product should have MTCTE Approval Certificate and the copy of the certificate to be submitted along with the bid.

39. Armoured Power cable

3 core, 2.5 sqmm armoured copper core power cable with standard specification

40. Floor mounted Rack 24U enclosure

1. Portable Rack
2. 800x600 mm depth or better
3. Rack with Front Glass Door & Castor Wheels
4. Numbers of Rack trays: 3
5. Number of Fan for Heat dissipation (nos): 4
6. 5/15-amp 6 socket PDU & Hardware accessories
7. Free-Standing mounting

8. Transparent Toughened Glass at front
9. Dust or water resistant
10. Vertical & Horizontal managers
11. With Secure Locks
12. Support cable entry from top or bottom
13. CRCA Steel or better material for rear doors
14. For Indoor Use

41. 9U Rack

1. 600 mm or more depth (as per site requirement)
2. Wall Mount
3. Front Toughened Glass with lock and keys
4. With PDU
5. For Indoor Use

42. Outdoor Vandal Proof box as per site requirement

1. As per site requirement
2. With proper gland as per site requirement Junction box is to accommodate all the equipment like Network switch, cable termination strip, and other components at the camera site. Hence, this shall be mounted firmly on the pole with GI clamps.
3. Ingress Protection rating shall be such that the equipment inside the box is protected from dust and water. (As per the condition of the location)
4. It shall be lockable and shall have all cable entries at the bottom sealed in such manner that it should give protection from dust and water entering through these entries.
5. All the equipment except the cameras shall be accommodated inside and is to be mounted on same pole.
6. For terminating all cables, it shall have suitable terminating blocker connections.
7. All the cables shall be clamped with suitable clamp to avoid the damage of the core/conductors due to cables weight.
8. A canopy is to be attached to the pole

43. CVT/ Voltage Stabilizer

1. Rating 5 KVA
2. Type of winding: Copper
3. Input Voltage Range: Single Phase Sine wave (90V-300V)
4. Overall Efficiency: 95%
5. Standard Specifications

44. MCB

Standard specification with ratings as per site requirement

45. Power Surge Protector

Lightning and Surge Protection for power line at Distribution level Between Power Line to Neutral, Between Neutral to Earth, Between to Phase to Neutral of class I/II/B/C.

46. Lightning arrestor with all accessories

Standard specification. Separate Earthing as well as separate copper cable of size 6 sqmm shall be used for grounding of lightning arrestor without additional cost to ECL

47. Earthing

1. Construction and Installation of Separate maintenance free Earthing of less than 2 Ohms
2. Chemical filled Earth electrode of suitable size should be used
3. Pipe in pipe one or more electrode, carbon-based ground enhancing material to bring the resistance less than 02 ohms.
4. The earthing shall be separated by other earthing by at least 10 feet.
5. The earth wire should be of same material as that of earth electrode used.

48. 1 KVA UPS with 60 mins backup

1. Rating 1 KVA/600 Watts or more Line Interactive/ Offline
2. Input Voltage Range: 160 V - 280 V, 50/60 Hz
3. Output Voltage: 230 +/- 10%
4. Transfer Time: 10 ms or lower
5. Indicator: On mains/ On Battery/ Low Battery
6. Alarm: Fault/Main Fail/Overload
7. Short circuit Protection
8. With all other accessories, required for installation & proper operation of UPS
9. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

49. 1 KVA online UPS with 120 min backup with isolation transformer (inbuilt / external)

1. Rating 1 KVA or more online Double Conversion UPS.
2. Waveform: Sinusoidal
3. Input Voltage Range: 160 V - 280 V, 50/60 Hz
4. Output Voltage: 230 +/- 10%
5. Output power factor 0.8
6. THD of input current: less than 10%
7. Overall efficiency: 80%
8. Back Up time: 120 mins or more on full load. Minimum 3120 Vah batteries
9. Operating Temperature: 0° to 45°C or more
10. Indicator: On mains/ On Battery/ Low Battery
11. Alarm: Mains fail, Low battery alarm, UPS warning, Overload, Fault & Bypass mode
12. Over Load and Short circuit Protection
13. With all other accessories, required for installation & proper operation of UPS
14. With Input power factor correction
15. With Inbuilt/ External Isolation Transformer
16. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

50. 3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer (inbuilt / external)

1. Rating 3 KVA or more online Double Conversion UPS.
2. Waveform: Sinusoidal
3. Input Voltage Range: 160 V - 280 V, 50/60 Hz
4. Output Voltage: 230 +/- 10%
5. Output power factor 0.8
6. THD of input current: less than 10%
7. Overall efficiency: 80%
8. Back Up time: 120 mins or more on full load. Minimum 6240 VAh batteries.
9. Operating Temperature: 0° to 45°C or more
10. Indicator: On mains/ On Battery/ Low Battery
11. Alarm: Mains fail, Low battery alarm, UPS warning, Overload, Fault & Bypass mode
12. Over Load and Short circuit Protection
13. With all other accessories, required for installation & proper operation of UPS
14. With Input power factor correction
15. With Inbuilt/ External Isolation Transformer
16. With Battery warranty of 03 years
17. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

51. CPE Radio

1. MIMO: 2X2 MIMO - OFDM
2. Mode of operation: can work as Subscriber Module/CPE radio for PMP topology and can work as PTP radio for PTP topology
3. Operating frequency band: ISM unlicensed free band
4. Channel Size: 20MHz, 40 MHz and 80MHz channel size
5. DUPLEX SCHEME: Time Division Duplex (TDD)
6. Up Link and Down Link Ratio: Configurable
7. Throughput: 200 Mbps
8. Ethernet Interface: 10 / 100 / 1000 BaseT, rate auto negotiated
9. Protocol: IPv4/IPv6 (Dual Stack), SNMP(v1/v2/v3), IGMP
10. VLAN: 802.1Q with 802.1p priority
11. Top Adaptive Modulation: 256QAM
12. Encryption: AES 128
13. Spectrum Analyzer: can show the noise level in frequency channels to help in selecting the best quality frequency channel
14. QoS: Three level priority with packet classification, MIR/CIR support
15. Operating temperature: 0°C to 50°C
16. Environment Compliant: IP 67 or more
17. Antenna Gain: minimum 24 dBi gain
18. Latency: Less than 20 ms
19. System standards: All the radio equipment shall comply fully with all customer defined mandatory requirement and conforming to the latest version of ETSI/ FCC / IEEE & must have ETA / TAC (Type Approval Certificate) approval of WPC (copy of the same shall be submitted along with the bid). The offered radios should be from trusted telecom approved OEM / certified from CERT-in or their empaneled organizations and the same may be submitted at the time of supply. The necessary documentary evidence in this

regard against the proof of status of application may be submitted from the OEM along with the bid. The offered radios should have MTCTE Approval Certificate and the copy of the certificate to be submitted along with the bid

20. Network Management Security, access control, report generation and debugging facility along with other standard features should be provided. The networking management system at AHQ (Network Operator Control) to manage and configure base stations and remote station unit. The functionality includes automatic and manual device discovering of all network elements such as base stations, remote stations etc and system shall keep details of all network elements. It should provide throughput / bandwidth test tools.
21. Other Requirements:
    a. With Surge Suppressor/Lightning Protector Unit (LPU) (for protection against lightning) near the device and arrangement for grounding the device
    b. With all accessories for mounting and installation of radio
22. It shall be IP V 6.0 compliant

52. Point to Multipoint Radios

1. MIMO: 2X2 MIMO – OFDM
2. Operating frequency band: ISM un licenced free band.
3. Channel Size: 20MHz, 40 MHz and 80MHz
4. Channel Size: channel configurable as with 20MHz, 40 MHz and 80MHz
5. Channel Selection: Automatic/ Dynamic Channel Selection
6. Up Link and Down Link Ratio: Configurable
7. Throughput: 600 Mbps or more
8. Ethernet Interface: POE 10 / 100 / 1000 BaseT, rate auto negotiated
9. Protocol: IPv4/IPV6, SNMP (v1/v2/v3), HTTPS
10. Spectrum Analyzer: can show the noise level in frequency channels to help in selecting the best quality frequency channel
11. Adaptive Modulation: BPSK / QPSK / 16QAM / 64QAM / 256QAM
12. Encryption: AES 128
13. Supports: Minimum 10 Subscriber Modules
14. GPS Synchronization: GPS synchronisation should be supported to streamline Radio TDD synchronisation of Intra & Inter-sites to mitigate mutual interference
15. QoS: Three level priority with packet classification
16. VLAN support: 802.1Q with 802.1p priority
17. Operating temperature: 0°C to 50°C
18. Environment Compliant: IP 67 or more
19. Antenna Coverage and Gain: 120-degree coverage with 14 dBi gain
20. System standards: All the radio equipment shall comply fully with all customer defined mandatory requirement and conforming to the latest version of ETSI/ FCC / IEEE & must have ETA / TAC (Type Approval Certificate) approval of WPC (copy of the same shall be submitted along with the bid). The offered radios should be from trusted telecom approved OEM / certified from CERT-in or their empaneled organizations and the same may be submitted at the time of supply. The necessary documentary evidence in this regard against the proof of status of application may be submitted from the OEM along with the bid. The offered radios should have MTCTE Approval Certificate and the copy of the certificate to be submitted along with the bid
21. Network Management Security, access control, report generation and debugging facility along with other standard features should be provided. The networking management

system at AHQ (Network Operator Control) to manage and configure base stations and remote station unit. The functionality includes automatic and manual device discovering of all network elements such as base stations, remote stations etc and system shall keep details of all network elements. It should provide throughput / bandwidth test tools.

22. Other Requirements:

a. With Surge Suppressor/Lightning Protector Unit (LPU) (for protection against lightning) near the device and arrangement for grounding the device

b. With all accessories for mounting and installation of radio "

23. It shall be IP V 6.0 compliant

53. PTP radio

1. MIMO: 2X2 MIMO - OFDM
2. Mode of operation: PTP radio for PTP topology
3. Operating frequency band: ISM unlicensed free band
4. Channel Size: 20MHz, 40 MHz and 80MHz channel size
5. DUPLEX SCHEME: Time Division Duplex (TDD)
6. Up Link and Down Link Ratio: Configurable
7. Throughput: 200 Mbps
8. Ethernet Interface: 10 / 100 / 1000 BaseT, rate auto negotiated
9. Protocol: IPv4/IPv6 (Dual Stack), SNMP(v1/v2/v3), IGMP
10. VLAN: 802.1Q with 802.1p priority
11. Top Adaptive Modulation: 256QAM
12. Encryption: AES 128
13. Spectrum Analyzer: can show the noise level in frequency channels to help in selecting the best quality frequency channel
14. QoS: Three level priority with packet classification, MIR/CIR support
15. Operating temperature: 0°C to 50°C
16. Environment Compliant: IP 67 or more
17. Antenna Gain: minimum 24 dBi gain
18. Latency: Less than 20 ms
19. System standards: All the radio equipment shall comply fully with all customer defined mandatory requirement and conforming to the latest version of ETSI/ FCC / IEEE & must have ETA / TAC (Type Approval Certificate) approval of WPC (copy of the same shall be submitted along with the bid). The offered radios should be from trusted telecom approved OEM / certified from CERT-in or their empaneled organizations and the same may be submitted at the time of supply. The necessary documentary evidence in this regard against the proof of status of application may be submitted from the OEM along with the bid. The offered radios should have MTCTE Approval Certificate and the copy of the certificate to be submitted along with the bid
20. Network Management Security, access control, report generation and debugging facility along with other standard features should be provided. The networking management system at AHQ (Network Operator Control) to manage and configure base stations and remote station unit. The functionality includes automatic and manual device discovering of all network elements such as base stations, remote stations etc and system shall keep details of all network elements. It should provide throughput / bandwidth test tools.
21. Other Requirements:

a. With Surge Suppressor/Lightning Protector Unit (LPU) (for protection against lightning) near the device and arrangement for grounding the device
b. With all accessories for mounting and installation of radio "

22. It shall be IP V 6.0 compliant

54. 10 m Tubular Swaged pole

1. Hot Dip Galvanized Swaged Tubular Poles as per IS:2713: 1980, with Cap and Base Plate
2. Shall be made of seamless or welded tubes of suitable lengths swaged and jointed together. No circumferential joints shall be permitted in the individual tube lengths of the poles. If welded tubes are used, they shall have one longitudinal weld seam only: and the longitudinal welds shall be staggered at each swaged joint.
3. Poles shall be well-finished, clean and free from harmful surface defects. Ends of the poles shall be cut square. Poles shall be straight, smooth and cylindrical.
4. The poles shall be Hot dip Galvanized (inside & outside) having minimum mass of zinc coating 400 gms per sq.mtr. as per IS 6745/1972.
5. The poles shall be marked with designation, manufacturer's identification, year of manufacture and name of the purchaser
6. Effective length of each section: Bottom: 5.2 m, Middle: 2.4 m, Top: 2.4 m
7. Joint Length: Bottom: 300 mm Top: 230 mm
8. Outside diameter and thickness of each section: Bottom: 139.7 mm X 4.50 mm, Middle: 114.3 mm X 4.50 mm, Top: 88.9 X 3.25 mm
9. Approximate weight of the Pole: 128 Kg (The mean weight for bulk supplies shall be within 92·5 percent of its calculated value. The weight of any single pole shall not fall below by more than 10 percent)
10. Thickness: 5 mm (or as per requirement of Pole length and weight)
11. With Civil Works
12. Length of The Pole: 10 Meters
13. With additional arrangements of stepper and with mounting fixture of radios
14. Test report to be submitted along with the supply

55. 27 Meters 3 Leg Self Supported Galvanized Towers

1. 3 leg self-supporting tower of height minimum 27 meters (or more as per site requirement) from the surface.
2. The tower shall comprise sections of suitable lengths (not exceeding 3 mtrs).
3. The material used shall be mild steel/HT steel as per IS 3601:2006 and foundation bolts shall be as per IS5624.
4. The towers shall be designed as per IS 875 for wind load calculation and IS 800 for design practices.
5. Wind load shall be considered based on area of tower structure and also ladder, safety rings, mounted equipment, platform etc.
6. All ferrous metal shall be 50-micron thick hot dip zinc galvanized (internally and externally).
7. The complete drawing details of the tower Erection & Foundation Diagram of Tower shall be submitted along with bid document
8. The tower foundation shall be designed based on the maximum wind speed experienced by the tower (180 Km/hr), and the weight of the tower and the safe bearing capacity (10 tons / sq. meter).
9. The foundation shall be of Reinforced Concrete and the foundation depth shall be 0.16 X total height of the tower. Backfilling should have well compacted.

10. Concrete foundation shall be prepared with a mixture of concrete, pebbles and sand in 1:1:2 [M25] ratios by the supplier.
11. Concrete mix grade shall be evaluated by subjecting the readymade concrete mix to cube test while making the foundation
12. The tower shall be fixed on the concrete platform with suitable bolts and these bolts shall be 150 micron thick galvanised.
13. It shall have a provision to mount a camera to one of the tower leg members.
14. It shall have a provision to mount a radio communication equipment on one of the tower leg members.
15. Tower shall have facility for climbing to the top.
16. Separate Lightening arrestor (in addition and shall be under the scope of the supply of the tower) made of copper shall be installed and mounted at the top with its connecting wire running to earth pit.
17. Separate maintenance free Earthing (in addition and shall be under the scope of the supply of the tower) of less than 2 Ohms shall be installed with the tower
18. Tin coated copper wire IS mark of gauge 50 sq. mm shall be installed for connecting lightning arrestor to the earthing.
19. Twin type aviation warning lights at the top level
20. Manufacture's identity marking and item marking shall be provided on each section by punching & painting for Enabling easy assembly and traceability. A plate indicating the manufacturer's identity, year of manufacture, wind velocity and the maximum load capacity of the tower shall be indicated for display at the bottom of the tower.

56. Long Range RFID Reader/Writer with pole and mounting

1. Operating Frequency: 865-867 MHz {or any other frequency (Regulatory approval from WPC, if applicable should be submitted along with offer)}
2. Ingress protection: IP 68 or better (Certificate from NABL accredited laboratory)
3. Operating range: Read up to 08 Meters and write up to 05 Meters (at least).
4. System should support both Read and write action on either hopping frequency or fixed frequency carrier as required
5. Air interface: EPC global Gen 2 UHF (ISO/IEC-18000-6C)/ ISO- 18000-6B
6. Communication interface: RJ45/ Serial port- RS232/ RS485/ Wiegand/ 12 pin screw terminal connector or any other interface as per system requirement
7. Network Protocol: - should support DHCP, TCP/IP, DNS, SNMP, SNTP, IPV4, IPV6 or as per system requirement
8. Should be Outdoor type, Rugged design to work well in mines environment, dusty & metallic environment
9. Should have built-in or external protection against lightening
10. Should have adaptability, compatible & easy to integrate with other systems present at the weighbridge
11. Interference elimination.
12. High data speed and anti-collision for multi tag reading
13. Advanced Anti-Jamming
14. RFID Reader should have built-in Antenna and should be compact.
15. The system should work well in dirty, electrically noisy, and metallic environments.
16. No polarization constraints. Capable of reading tags in any orientation
17. All equipment should be outdoor type.

18. Compliance: Equipment Type Approval of the Quoted make and model as per existing norms of WPC wing Ministry of Communications and Information Technology, Department of Telecommunications. Necessary documentary evidence may be submitted along with the bid.
19. BIS Certificate for the quoted make and model of RFID reader writer. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
20. The MAC address of RFID Reader (outdoor) must be registered in the name of OEM supplying RFID Reader (outdoor). OEM Self Declaration is required in this regard
21. Operating Temperature Range: 0º to 55º Centigrade or better
22. Humidity (non-condensing): should be able to work at 90% RH maximum.
23. Mounting: Pole Mounting/ Tower Mounting
24. Tag Buffer: More than 25000 RF Tags or better
25. Minimum Tag reading Capacity: 25 or more Tag/Sec
26. Minimum Tag writing Capacity: 05 or more Tag/Sec
27. RF Power Output: up to 30 dBm, Software adjustable
28. The OEM Must provide in RF Test Report of the Reader and the same must confirm the name of OEM as quoted by the bidder
29. The Reader must be tested for: IEC-61000-4-6 and IEC-61000-4-5 or Equivalent EN/ UL Standards
30. All other necessary hardware, software & accessories as required for making system live
31. With supply of separate POLE, without additional cost to ECL, for RFID, Traffic light Signalling, mounting of CCTV cameras and flood light:
    a. Length of Pole: Min 6 Mtr or above as per site requirement
    b. Material: GI Powdered coated / Mild steel
    c. Size: Diameter: 15cm (or as per requirement of Pole length)
    d. Thickness: 5mm (or as per requirement of Pole length)
    e. Pole is required to be painted with rust free painting periodically
    f. Wind Speed: It can withstand Max wind speed at sites that are located in West Bengal, Jharkhand
32. Cable Wiring Provision internally with required protection to avoid any damage

57. Signal Light for Traffic control

Traffic Signal (two LED Traffic Lights one red & one green) with standard specifications to be mounted to Pole at entry and exit of check post. All accessories including mounting pole with High Strength

58. 5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor

1. The length of the Boom Barrier should be 5 meter or above
2. Automatically triggered/Open after getting information from Local Workstation/Central server/ RFID reader about authenticity of vehicle. However, bypass system for manual operation when no power should be made available or in any other emergency situation/cases
3. Safety arrangement for preventing boom from closing onto the vehicle shall be provided using suitable mechanism involving sensors. The safety sensors should be high enough to prevent accidental dropping of Boom-Barrier during passage of cattle or humans.
4. All the housing for internal parts should be of rust and corrosion free metal or alloy of high strength with suitable epoxy coating & provided with radium reflector tape

5. Boom Barrier shall be BIS/CE certified
6. Degree of protection – IP 54 or better.
7. Operating Temperature: 0 °C to + 55 °C or better
8. Humidity: 5 % to 95 % (non-condensing)
9. Communication: Relay interface to connect any Access Control or R.F identification reader device and shall have RJ-45 for LAN Communication, RS-232 or RS-485 (For Data) or USB to fetch data at Site Location
10. Power input should be single phase 230 V AV + 10 % 50 Hz or as per system requirement
11. The drive mechanism should include a torque motor with a gearbox, with or without a counterweight. However, the control mechanism must be adjustable to ensure proper damping during the opening and closing of the boom barrier
12. Protective shock resistant bumper made of rubber should also be provided to protect the barrier
13. Common entry-exit points exist in many check posts of ECL in which single boom barrier with two Readers will be installed for movement of vehicles through same gate. At such points, the Boom-barrier should be capable of being triggered by both the RFID readers. The successful bidder has to arrange for necessary spares (network card, controller card), if required, and implement such a mechanism, without extra cost to ECL
14. The Boom Barrier should have and operate with Photo Electric Loop Detector or any other sensor-based loop detector, which may be mounted on poles in pairs (one transmitter and one receiver) at both sides of the road along with civil foundation and underground wiring through GI Pipes as required. Alternatively, any other type of sensor may be installed on the approach road itself or as suitable for operation. A TCP/IP based controller should also be installed for communication. The system should work to ensure proper inputs as required for smooth and hassle-free operation of Boom Barrier.
15. The successful bidder is required to complete all the necessary civil work for installation of barrier, boom rests and safety sensors etc. The foundation must be of concrete, and it should be at a minimum height of 12" above the ground. All wiring must be laid underground through proper conduits
16. The Boom Barrier Controller should be Tested for IEC 61000-4-5 for Surge Immunity
17. Speed of barrier should be programable while installing as per frequency of traffic.
18. Opening Time and Closing Time of Boom shall be less than or equal to 6 sec.
19. A TCP/IP based controller for communication, Backup and restore database, shall be provided which shall support up to two boom barrier, two RFID Readers. Timed Anti-pass back; Tag-in/Tag out; Tag-in/Free out.

59. LED Flood Light along with all accessories

1. LED Rating (in Watts)/System Wattage/Rated Power: 240 Watt or better
2. Input operating Voltage range and frequency: 90 to 300 Volts AC at 50 Hz+/-2 or better
3. Automatic Higher Cut off voltage above 300 volt
4. Driver Efficiency Upto 100W: ≥ 90% or better
5. LED chip Efficacy ≥ 130 Lm/Watt or better
6. Should have short circuit Protection; Over load protection; Over Voltage protection; Reverse polarity;
7. Insulation resistance between earth and current carrying part: >100 M Ω
8. Ingress Protection (IP Rating): IP66 or better
9. Impact Protection: IK-08 or better

10. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.

60. Portable Cabins 10' x 10' × 8' (LxBxH) with AC, Table, chairs, Fans, Lights, 2 windows with electricals, all accessories; foundation & civil work including spares

1. **Cabin:**
   a. One Room with two windows and roof in One way slope. Room Size 10' x 10' × 8' (LxBxH) (internal measurement) with Almirah, Split AC, Table, chairs (02 nos.), Fan, Lights (03 nos.), 2 windows with electricals, all accessories; foundation, paint & civil work
   b. Logo of ECL of size as per requirement to be painted
   c. The window panel material shall be Board Aluminium frame sliding shutter with plain glass.
   d. One Ventilator made out of Aluminium adjustable Louvers type channel and to be fixed in wall panel with suitable fasteners.
   e. With Erection and Installation of power cables, power points of 5 Amps and 16 Amps and other electrical accessories, casings, lights, fan, AC, stabilizer etc. as per the system requirement. No additional cost shall be paid by ECL for the supply and installation of electrical accessories.
   f. Any fittings or items which may not be specially mentioned in the specifications but which are necessary for commissioning of Cabin are to be provided by the Seller without any extra charge
   g. Insulation core of panel to be formed in one piece and provides desired structural and physical properties
   h. Internal face of wall of 8 mm thick and external face of wall of 9 mm heavy duty fire and water proof of cement fiber board with 30 mm high density thermocal sandwich between Internal and external walls or better
   i. Door panel Waterproof BWP flush door 34 mm thick or better
   j. Locking Door and Windows to prevent unwanted intrusion and theft.
   k. Floor Panel material: "at least 15 mm heavy duty cement fibre board inside and outside texture paint" or "at least 19 mm thick marine grade plywood confirming to IS 710 / latest to be laid on top and secured by self - tapping screws to the base members - Covered with 2 mm thick PVC sheets"
   l. All replacements / repairs shall be done in respect of fittings and fixtures during contract period.

2. **Chairs:** 02 chairs, Low/Mid back revolving chair with Armrest, Tilt mechanism, Locking mechanism, Seat depth adjustment, Height adjustments, seat height 400 mm to 500 mm, Width of seat 400 mm to 500 mm, Depth of seat 400 mm to 500 mm

3. **Table:** Wood thickness is 18mm overall with 2 mm edge-banding or better Dimension: Overall Height: 120 cm or better Overall width: 60 cm or better Overall Depth :75 cm or better

4. **Split AC:** Capacity: 1.5 Ton, Split, 5 Star, High EER Rotary compressor, Evaporator and Condenser material copper, Panel Display: LED, Anti – Bacterial filer, Selectable Swing.

Separate double pole MCB with suitable current rating shall be installed with separate wiring, plug points for AC installation

5. **<u>Stabilizer:</u>** Rated KVA: 5 KVA, Input Single phase sine wave 160 to 270 V, Output Single Sine wave 220V, Copper Winding, High Voltage Cut off protection Built in Thermal Overload Protection - Protects the stabilizer and compressors during high temperature burnout, Wall Mounted

61. Handheld RFID reader along with all hardware & software accessories

1. Operating Frequency: 865~867 MHz {or any other frequency (Regulatory approval from WPC, if applicable should be submitted along with offer)}
2. Operating range: Read up to 15m
3. Communication interface: GPRS (4G, 3G/2G), Wi-Fi (IEEE 802.11 a/b/g/n or better), Bluetooth 5.0
4. Build-In rechargeable battery with continuous use over 8hrs
5. Should able to decode IS0-18000B18000-6B/ ISO 18000-6C, EPC
6. Application software & other necessary accessories should be provided
7. Should have built-in antenna
8. Capacitive Touch display and scan key buttons
9. Sound / LED indicators
10. Protection: IP65 or better
11. Optional SD card support
12. Product should be certified for relevant BIS standard from any BIS/NABL recognized/ accredited laboratory. Necessary documentary evidence may be submitted along with the bid.
13. Compliance: Equipment Type Approval of the Quoted make and model as per existing norms of WPC wing Ministry of Communications and Information Technology, Department of Telecommunications. Necessary documentary evidence may be submitted along with the bid.

62. RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)

RFID CARDS- UHF RFID CARDS IN HARD PLASTIC WITH ONE SIDED PRINTING.

1. Operating Frequency: 865~867 MHz{or any other frequency (Regulatory approval from WPC, if applicable should be submitted along with offer)}
2. Protocol: EPC CLASS 1 Gen 2/ ISO 18000-6C/ ISO- 18000-6B
3. Memory capacity: 512 bits or higher
4. Read/Write type
5. Material: PVC/PET or better encapsulation to maintain the integrity of the tag in harsh environment
6. Tag ID should not be edited or reprogrammed. It should have factory programmed worldwide unique ID.
7. Operating mode: - Fixed frequency or FHSS software programmable
8. Operating temperature: 0 °C to + 60 °C or better
9. Should be Outdoor type, work well in mines environment & have rugged design
10. Ingress protection: - IP 68 or better

11. Reusable with minimum 150000 cycles
12. Flexible read range: up to 8 mtr
13. RFID tags is a tamper proof tag and can be easily attached with self-adhesive tape
14. Use: Cards in hard plastic with one sided printing and can be easily attached with self-adhesive tape.
15. All the tags that will be supplied must have a unique serial no printed on them along with ECL logo.

63. RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)

RFID TAGS - TAMPERPROOF WINDSHIELD PASSIVE UHF RFID TAGS WITH ONE SIDED PRINTING.

1. Operating Frequency: 865~867 MHz{or any other frequency (Regulatory approval from WPC, if applicable should be submitted along with offer)}
2. Protocol: EPC CLASS 1 Gen 2/ ISO 18000-6C/ ISO- 18000-6B
3. Memory capacity: 512 bits or higher
4. Read/Write type
5. Material: PVC/PET or better encapsulation to maintain the integrity of the tag in harsh environment
6. Tag ID should not be edited or reprogrammed. It should have factory programmed worldwide unique ID.
7. Operating mode: - Fixed frequency or FHSS software programmable
8. Operating temperature: 0 °C to + 60 °C or better
9. Should be Outdoor type, work well in mines environment & have rugged design
10. Ingress protection: - IP 68 or better
11. Reusable with minimum 150000 cycles
12. Flexible read range: up to 8 mtr
13. RFID tags is a tamper proof tag and can be easily attached with self-adhesive tape
14. Use: Tamperproof windshield passive UHF RFID tags with one sided Printing and can be easily attached with self-adhesive tape.
15. All the tags that will be supplied must have a unique serial no printed on them along with ECL logo.

64. Reflective acrylic sticker

1. Yellow-coloured radium acrylic sticker of size (4" X 16") with 4” large size black-coloured standardized font displaying the vehicle registration number needs to be affixed on top of the windshield so that the clear view of the vehicle registration number can be recorded through CCTV camera or through ANPR system installed at weighbridges and entry / exit barriers.
2. It will be bidder’s responsibility to paste Vehicle registration number sticker/paint Vehicle registration number (as per ECL requirement time to time) on vehicles.

65. Truck Centering System with controller and all accessories

1. The system must detect if a vehicle is properly positioned on the weighbridge platform and identify multiple vehicles simultaneously.

2. The sensors can be IR/ultrasonic/photo/magnetic loop or any other proven technology with Sensing range 5 meters or higher
3. Indoor unit and sensors enclosure grade must be IP65
4. Input power supply must be 180-240V, 50Hz
5. Should have surge and lightning protection.
6. Controller must have RS232/ USB/ TCP-IP ports for PC connection and operate traffic signal lights to indicate correct and incorrect vehicle positioning
7. Controller must issue a command and the weight must only be recorded if and only if the vehicle is properly centered on the platform.
8. Must have provisions for both light and sound alarms.
9. Sensors should be mounted on ISI-marked MS poles (2.5" or more) at least 2 feet from the weighbridge, with a height of 5 feet above the platform. Poles must be set in a concrete foundation, and all wiring must be concealed in underground conduits

66. AUDIO SYSTEM ALONG WITH ALL HARDWARE & SOFTWARE ACCESSORIES

1. To be used along with signal lights for proper centering of the vehicle on the platform by guiding the vehicle operator to move forward or backward during centering of the vehicle.
2. Must have amplifier, speakers and mic for announcement and guiding the drivers to center their vehicle on the platform. The same should be automated through controller and must also allow for manual override by the operator if needed.
3. Interface: As per system requirement
4. Can be easily integrated with application
5. Should be Audible in Coal Mining environment